

Uncertainty Quantification and Calibration in Template-Based Retrosynthesis Models: A Cross-Architecture Study of LocalRetro and MHNreact

Yamen Fargaly^{*1}

¹Princeton University

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Abstract

Template-based single-step retrosynthesis models output a softmax distribution over a template vocabulary, and their top score is routinely read as a confidence. We ask whether that built-in signal reliably separates correct from incorrect predictions, and whether the separation survives distribution shift toward rare and novel templates. Across two architecturally distinct models on USPTO-50K ($n = 5,007$), LocalRetro (a graph neural network) and MHNreact (a Modern Hopfield retrieval network), the answer is no, and the failure takes two distinct forms. LocalRetro is well calibrated in aggregate but collapses sharply under template rarity; MHNreact is more uniformly miscalibrated. The standard perturbation remedies, MC dropout and deep ensembling, do not rescue the rare tail in either model, and in MHNreact both signals are inverted rather than merely uninformative, a boundary-robust and seed-replicated result. Per-tier (Mondrian) conformal selective prediction is the one method that restores error control. Finally, a held-out-reaction-class experiment confirms the collapse under genuine novelty: with two entire reaction classes removed from training, top-1 accuracy falls to 0.00–0.25% while the model remains confident (out-of-distribution ECE up to 0.39), the deployment failure mode that matters most.

Keywords

uncertainty quantification; calibration; retrosynthesis prediction; template-based models; conformal prediction; deep learning; distribution shift; graph neural networks

^{*}Corresponding author. Email: yf0550@princeton.edu. ORCID: <https://orcid.org/0009-0000-5289-6265>

1 Introduction

This paper presents a cross-architecture uncertainty quantification (UQ) and calibration study of template-based retrosynthesis models. It combines an end-to-end analysis of LocalRetro, a graph neural network for single-step retrosynthesis prediction,¹ with a second, architecturally distinct model, MHNreact,² a Modern Hopfield Network operating on molecular fingerprints. LocalRetro combines local atom and bond reactivity scoring with template based reactant prediction; MHNreact instead encodes molecules as fingerprints and retrieves templates via Hopfield association. Both output a softmax distribution over a template vocabulary, which is what makes the calibration and OOD-frequency methodology transferable between them.

The question we set out to answer: does a template-based model’s built in confidence signal ($1 - \text{top_score}$) reliably separate correct from incorrect predictions, and does that separation survive under distribution shift toward rare, novel templates? The headline result is that it does not: **confidence calibration collapses under template rarity in both models**, with two distinct failure signatures (LocalRetro suffers a sharp distribution-shift collapse; MHNreact is more uniformly miscalibrated). The obvious perturbation-based fixes, MC dropout and deep ensembling, do not rescue the rare tail in either model, and in MHNreact both perturbation signals are not merely uninformative but *inverted*, a boundary-robust, seed-replicated finding. What does work is per-tier (Mondrian) conformal selective prediction, an architecture-agnostic fix that correctly abstains where no threshold can deliver a reliable guarantee.

Every result below comes from a single trained checkpoint per model, with template extraction, preprocessing, and all downstream analyses run against that fixed checkpoint (see Section 3). Figures and tables presenting the full result in both architectures are in the main text; single-architecture detail, distribution plots, and ROC curves that support (without adding to) the main-text AUROC numbers are collected in the Supplementary Information at the end of this document.

Contributions

1. **First calibration-under-shift study for template-based retrosynthesis.** We characterize how the built-in confidence signal of single-step template models behaves under template-frequency shift, across two architectures (GNN and Modern Hopfield retrieval) on a common benchmark.
2. **Two failure signatures, not one.** We show the miscalibration is architecture-dependent: a sharp distribution-shift collapse in LocalRetro versus uniform miscalibration in MHNreact, distinguishing model-specific from general behavior.
3. **A negative result on perturbation UQ.** MC dropout and deep ensembling do not rescue the rare tail in either model, and are inverted in MHNreact. We harden this with a boundary sweep, a second independent ensemble, and a named benign mechanism (multi-valid-template ties), so it is reported as a robust finding rather than an artifact.
4. **A working mitigation and a genuine-novelty confirmation.** Mondrian conformal selective prediction restores per-tier error control, and a held-out-reaction-class experiment reproduces the confident-but-wrong collapse under true out-of-distribution novelty (0% accuracy at substantial confidence), moving the result from suggestive diagnostic to demonstrated phenomenon.

Key Findings

- Baseline confidence ($1 - \text{top_score}$) separates correct from incorrect predictions with AUROC = 0.710 (LocalRetro) and 0.738 (MHNreact), a real and statistically significant effect in both models (LocalRetro: Spearman $\rho = 0.364$, $p < 10^{-150}$).
- **The most consequential finding: calibration degrades with template rarity in both models.** Using rank-based vocabulary-matched tiers (exact same n per tier for both models), both show highest calibration error in the rare tail, though with different profiles: the rare/high ratio is $3.2\times$ vs $2.9\times$, but the rare/mid ratio is $7.3\times$ vs $1.8\times$, reflecting genuinely different failure shapes. LocalRetro collapses sharply under novelty (debaised ECE 0.024 to 0.291); MHNreact (retrained with verified early stopping, Section 3.1) shows degradation concentrated in the tail (0.038 to 0.107 on absolute-count tiers), with

a clean monotonic gradient on the rank-based tiers. In LocalRetro’s very-rare tier ($n=100$), confidence discrimination is statistically indistinguishable from random (AUROC 0.464, 95% CI [0.248, 0.706]).

- MC dropout (20 stochastic passes, 4 dropout layers) does not improve on baseline for LocalRetro. Prediction variance across passes (`std_top_score`) is a weak signal on its own for LocalRetro (AUROC = 0.545), and no combination we tried beat the simple softmax confidence. MHNreact’s MC dropout variance behaves very differently: it is not merely weak but inverted (AUROC = 0.330, overall): higher dropout variance predicts a *correct* prediction more often than an incorrect one, and this holds up under a boundary-robustness sweep (Section 7).
- A 4-seed deep ensemble improves overall discrimination for LocalRetro (paired DeLong test, $\text{diff} = 0.017$, $p = 3.6 \times 10^{-8}$) but not for MHNreact ($\text{diff} = 0.001$, $p = 0.868$). Worse, MHNreact’s ensemble disagreement is inverted in the very-rare tier (AUROC = 0.427, Bonferroni CI [0.394, 0.461]), the same direction of surprise as the MC dropout inversion, and this replicates on a second, completely independent 4-seed ensemble (AUROC = 0.4277) and across every very-rare-tier boundary tested.
- Mondrian (per-tier) conformal prediction is the proposed fix, and is architecture-agnostic. A global threshold catastrophically fails the very-rare tier (keeps 9% of predictions with a 100% error rate). A per-tier Mondrian threshold correctly abstains on the very-rare tier and holds the error target in other tiers, for both models. A follow-up Hoeffding-UCB-corrected threshold shows the naive version has an optimism bias in MHNreact’s smaller tiers; the corrected version is honest but often equivalent to abstaining until more calibration data is available.
- **A genuine held-out-reaction-class experiment (Section 6) confirms the frequency-proxy collapse was, if anything, an understatement.** On truly novel reaction classes, LocalRetro scores 0/909 and 0/811 correct across two independent classes (architecturally impossible, since the held-out templates do not exist in its retrained vocabulary) and MHNreact scores 1/909 and 2/811 (astronomically unlikely but not architecturally forbidden, since its vocabulary is fixed upfront). Both models remain confidently wrong (OOD ECE 0.376/0.390 for LocalRetro, 0.268/0.257 for MHNreact), the practical failure mode that matters most for deployment.

2 Related Work

Calibration and uncertainty in neural networks. Modern classifiers are frequently miscalibrated,³ and the standard remedies are temperature scaling, deep ensembles,⁴ and MC-dropout Bayesian approximation.⁵ Calibration is usually measured with expected calibration error.⁶ Our work applies this toolkit to a domain where it has not been characterized and reports where the perturbation-based members of it fail.

Conformal and selective prediction. Conformal prediction gives distribution-free coverage guarantees,^{7,8} and Mondrian/class-conditional variants provide per-group control.⁹ We use Mondrian conformal selective prediction as the mitigation that works when perturbation UQ does not, and analyze where finite rare-tier sample sizes bound it.

Uncertainty in molecular and reaction models. UQ has been benchmarked for molecular property prediction (Hirschfeld et al.;¹⁰ Heid et al.¹¹) and for calibration and generalizability of probabilistic models on small chemical datasets (Tom et al.¹²), and for reaction-level prediction such as yield forecasting (Kwon et al.¹³). The closest prior calibration work on reaction outcomes is Schwaller et al.,¹⁴ which calibrates a template-free transformer for *forward* reaction prediction. To our knowledge no prior work studies calibration of template-based *single-step retrosynthesis* models under template-frequency distribution shift, nor compares this behavior across architecturally distinct models, which is the gap this study fills.

3 Methodology

Sections 3.1–3.3 describe the LocalRetro and MHNreact training pipelines; Sections 3.4–3.6 describe the shared OOD-frequency proxy, calibration metrics, and UQ signal protocols applied identically to both models.

3.1 Model and Data Pipeline

LocalRetro was trained from scratch on the standard USPTO-50K dataset (Schneider/GLN split: 40,009 train, 5,002 validation, 5,008 test reactions), using the official preprocessing pipeline (`Extract_from_train_data.py`, then `Run_preprocessing.py`, then `Train.py`). Template extraction and training used the same fixed checkpoint throughout all downstream analyses. Template extraction yielded 124 atom level templates and 547 bond level templates. Training used early stopping (patience = 5) and converged at epoch 19 of 50, with validation loss 0.0324 and test loss 0.0319. Top 1 accuracy on the full standard test set came out to 53.11% (2,659 of 5,007 correct), consistent with LocalRetro’s published stereo unaware top 1 accuracy range.

MHNreact² was trained from scratch on the same USPTO-50K standard split (Morgan fingerprint size 4096, RDKit template fingerprints, Hopfield association dimension 512) using the official training script. Training used early stopping on validation loss (patience 5), converging at epoch 6 with validation loss 3.193; this stopping point was independently verified by rerunning with patience 10, which found no improvement over the remaining 10 epochs and converged to the identical checkpoint. Top-1 accuracy on the test set: 46.7%. Correctness labels were obtained by applying each predicted template to the product SMILES via `rdchiral` and comparing the resulting reactants to the ground truth using RDKit canonical SMILES with fragment-order-independent comparison. Template frequency was computed from MHNreact’s own training data using its 11,800-template vocabulary, and the same four frequency tiers were applied.

3.2 Baseline UQ Calibration

For each test molecule, $UQ = 1 - \text{top_score}$, where `top_score` is the softmax probability assigned to the model’s top 1 predicted template (for LocalRetro, the (template, site) edit). Correctness (error = 0 or 1) was determined by canonicalizing the predicted top 1 reactant SMILES (RDKit, fragment order independent) and comparing it against the ground truth reactants.

3.3 MC Dropout

For LocalRetro, all `nn.Dropout` submodules (4 found in the trained model) were turned back on at inference while the rest of the model stayed in eval mode. For MHNreact, its `nn.Dropout` submodules (mol and template encoders, dropout $p = 0.2$) were reactivated the same way, verified against the model’s actual source rather than assumed. In both cases we ran 20 stochastic forward passes per molecule, producing `mean_top_score` (averaged prediction confidence), `std_top_score` (variance across passes), `agreement_rate` (fraction of passes agreeing with the consensus top 1 prediction), and `predictive_entropy`.

3.4 Out of Distribution Proxy: Template Training Frequency

We did not run a full held out reaction class retrain, since no reaction class labels were available for this dataset instance. Instead, each test reaction’s true template’s frequency in the training set served as a continuous proxy for how in distribution or out of distribution that reaction is. Templates seen thousands of times in training are about as in distribution as it gets; templates seen once or twice are functionally rare and OOD like without requiring any retraining. Test molecules were bucketed into four absolute-count frequency tiers (1 to 5, 6 to 20, 21 to 100, 100+ training occurrences) for within-model analysis. Because LocalRetro (671 templates) and MHNreact (11,800 templates) have very different vocabulary sizes, absolute-count tiers assign very different fractions of each test set to the same nominal tier, so cross-model comparison instead uses **rank-based** quantile cutting: each model’s test set is sorted by its own template’s training frequency and cut by rank position, forcing each tier to hold exactly the same number of molecules (n=500, 1,152, 1,652, 1,703) in both models regardless of vocabulary size or value ties.

3.5 Calibration Metrics: Reliability Diagrams, ECE, and Conformal Prediction

All AUROC values reported measure *discrimination* (does the signal rank correct predictions above incorrect ones?), not *calibration* (does stated confidence match observed accuracy?). We supplement discrimination results with: (1) reliability diagrams and ECE/MCE, both overall and per frequency tier; (2) equal-mass

(quantile) binning alongside standard equal-width binning, since fixed-bin ECE has a positive small-sample bias at small n that would artificially inflate the rare-tier vs. very-common-tier comparison; and (3) a simulated bias floor (500 replicates of perfectly-calibrated synthetic data at each tier’s actual confidence distribution and n) to report a debiased ECE. All AUROC values carry bootstrap 95% CIs (2,000 resamples). For rare-tier multi-signal comparisons, we additionally report Bonferroni-corrected CIs across the signals compared within that tier.

3.6 Mondrian Conformal Selective Prediction

We implement Mondrian (frequency-stratified) conformal selective prediction as a proposed fix. Standard (marginal) conformal sets a single score threshold targeting a global error rate; Mondrian sets a separate threshold per frequency tier, restoring the coverage guarantee within each tier independently. The combined nonconformity score is a logistic combination of $1 - \text{top_score}$ and std_top_score , fit on a held-out calibration half-split. Full results, including a Hoeffding upper-confidence-bound correction for finite-sample optimism, are in Section 8.

4 Baseline Confidence: A Moderate but Real Discriminator (Both Models)

The softmax derived confidence signal shows a real, statistically significant ability to separate correct from incorrect predictions in both models. LocalRetro’s overall ECE (0.051) is noticeably lower than MHNreact’s (0.087), though both mask substantial per-tier variation, which is the subject of Section 5.

Metric	LocalRetro (GNN)	MHNreact (Hopfield)
Top 1 Accuracy	53.11% (2,659 / 5,007)	46.69% (2,338 / 5,007)
AUROC (1 – top_score vs. error)	0.7104 [0.696, 0.724]	0.7376 [0.724, 0.752]
Spearman ρ	0.3637 ($p = 2.04 \times 10^{-156}$)	0.4107 ($p = 4.48 \times 10^{-203}$)
ECE (overall, 10 bins)	0.051	0.087
MCE (overall)	0.214	0.154
Mean UQ correct predictions	0.367	0.381
Mean UQ incorrect predictions	0.534	0.627

Table 1: Baseline UQ performance, both models (MHNreact retrained with verified early stopping, Section 3.1). Both signals discriminate correct from incorrect predictions; MHNreact’s higher Spearman ρ but higher overall ECE reflects the two-failure-signatures pattern developed in Section 5: MHNreact separates classes slightly better in rank but is less well calibrated in absolute probability terms.

This is setup, not the contribution: both models show a moderate, real, but unremarkable discrimination signal in-distribution. The interesting behavior, where that signal survives and where it fails, is the subject of the rest of this paper. Supporting figures (mean UQ by correctness, overall reliability diagram) are in Supplementary Information Section 10.1.

5 Calibration Collapses Under Template Rarity (Both Models)

This is the central finding of the paper. Both expected directional correlations hold in both models: rarer training templates correlate with more errors and with lower confidence, but the magnitudes are starkly mismatched between how fast accuracy collapses and how much confidence actually adjusts. The two models collapse in visibly different ways.

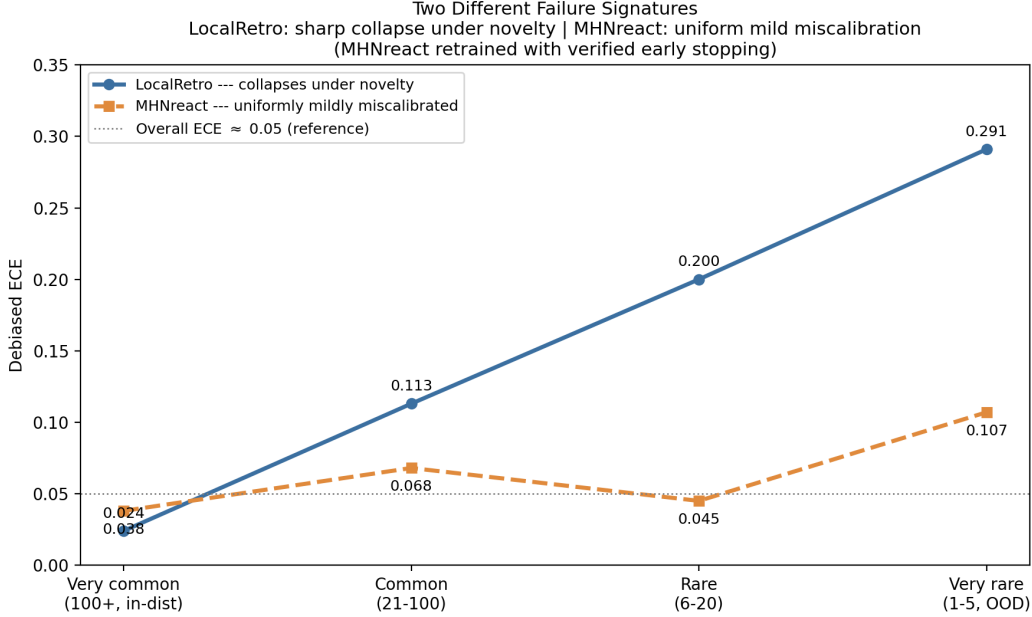


Figure 1: Two different failure signatures. LocalRetro is well-calibrated in-distribution (debiased ECE 0.024 at very common tier) and collapses sharply under novelty (0.291 at very rare). MHNreact (retrained with verified early stopping) is mildly but uniformly miscalibrated across all tiers (0.038–0.107), with no sharp collapse. Both patterns are well above the ideal (ECE = 0); they represent distinct miscalibration regimes, not one model being simply better than the other.

The contrast:

- **LocalRetro:** well-calibrated in-distribution (debiased ECE = 0.024 for very common templates), collapses sharply under novelty (0.291 for very rare). A model that is trustworthy in familiar territory and unreliable under distribution shift. Across the four absolute-count frequency tiers, accuracy collapses from 57.3% (templates seen 100+ times) to just 10.0% (templates seen 1 to 5 times), a 47.3 point drop, while mean UQ rises by only 0.110 over the same range. In the very-rare tier ($n=100$), confidence discrimination is statistically indistinguishable from random (AUROC 0.464, 95% CI [0.248, 0.706]).
- **MHNreact:** mildly but uniformly miscalibrated (debiased ECE ranging 0.038–0.107, no sharp collapse; retrained with verified early stopping, Section 3.1). A model with a more diffuse, architecture-wide calibration offset. MHNreact’s rare-tier debiased ECE (0.107) remains lower than LocalRetro’s *common-tier* ECE (0.113), though the margin has narrowed from the pre-retraining comparison (0.096 vs. 0.113). Unlike LocalRetro, no MHNreact tier’s confidence-discrimination CI comes close to spanning random performance; MHNreact retains real discrimination even in its very-rare tier (AUROC 0.634 [0.608, 0.661]).

Both models show their worst calibration in the rare tail; the shared degradation direction is the cross-architecture finding. The specific failure mode differs: LocalRetro suffers a distribution-shift collapse; MHNreact suffers a more uniform overconfidence. Mondrian conformal selective prediction (Section 8) addresses both, since it is agnostic to the shape of the ECE profile, requiring only the softmax confidence score and the template frequency label. Per-tier reliability diagrams for both models are in Supplementary Information

Rank-Based Vocabulary-Matched Confirmation

The two models use very different template vocabularies (LocalRetro: 671 templates; MHNreact: 11,800 templates), so absolute-count frequency tiers assign different fractions of the test set to each tier: LocalRetro’s very-rare tier (≤ 5 training occurrences) contains 100 molecules (2% of test set); MHNreact’s contains 1,835 molecules (36.6%). These are not the same population of reactions viewed through two models; they are different slicings, and a naive value-based quantile cut also fails to produce matched tiers because thousands of MHNreact templates share frequency=1. Rank-based quantile cutting (sort each model’s test set by its template’s training frequency, then cut by rank position) forces each tier to hold exactly the target fraction regardless of ties, giving four tiers of $n=500, 1,152, 1,652, 1,703$, identical for both models.

Tier (by rank)	LocalRetro (GNN)				MHNreact (Hopfield)			
	n	Acc	ECE deb	AUROC	n	Acc	ECE deb	AUROC
Rare (bot 10%)	500	0.252	0.197	0.642 [0.579, 0.702]	500	0.148	0.134	0.520 [0.456, 0.588]
Low (10–33%)	1,152	0.373	0.135	0.712 [0.680, 0.743]	1,152	0.309	0.093	0.644 [0.611, 0.677]
Mid (33–66%)	1,652	0.583	0.027	0.706 [0.682, 0.732]	1,652	0.490	0.076	0.708 [0.681, 0.733]
High (top 33%)	1,703	0.669	0.061	0.699 [0.671, 0.726]	1,703	0.645	0.047	0.741 [0.717, 0.764]
rare/high ratio			3.2×				2.9×	
rare/mid ratio			7.3×				1.8×	

Table 2: Rank-based vocabulary-matched tier comparison (MHNreact retrained with verified early stopping; Section 3.1). Each tier holds exactly the same number of molecules for both models. The rare/high ratios ($3.2\times$ vs $2.9\times$) are similar, but the rare/mid ratios ($7.3\times$ vs $1.8\times$) differ substantially, reflecting different ECE profiles across the frequency spectrum. All AUROC values carry 95% bootstrap CIs (2,000 resamples). LocalRetro’s debiased ECE series is not strictly monotonic (the high tier exceeds the mid tier); MHNreact’s is now strictly monotonic following retraining.

MHNreact was retrained from scratch with early stopping on validation loss (patience 5, converged at epoch 6, validation loss 3.193), achieving 46.7% top-1 accuracy, essentially unchanged from an earlier, methodologically flawed fixed-10-epoch run (46.6%) that had saved its final epoch rather than its best validation checkpoint. This retraining, and the confirmation that it did not materially change any result reported here, is detailed in Section 10. The rank-matched ECE/AUROC comparison is shown visually in Supplementary Information Section 10.2.

A Single Global Temperature Cannot Fix a Tier-Dependent Problem

Before testing perturbation-based fixes (Section 7), we test the simplest and most standard recalibration remedy: temperature scaling,³ a single scalar T fit to minimize calibration error, applied post-hoc to the model’s confidence. Since only the top-1 softmax probability is available (not the full logit vector), we use the standard one-vs-rest approximation: T is fit on a held-out calibration half-split to minimize negative log-likelihood of $p^{1/T}/(p^{1/T} + (1-p)^{1/T})$ against correctness, then applied to the test half.

The overall ECE improvement looks substantial for both models, but the per-tier breakdown shows why that is misleading (shown visually in Supplementary Information Section 10.4). For LocalRetro, essentially all of the overall gain comes from the very-common tier (85% of the data); the very-rare and rare tiers barely move ($0.408 \rightarrow 0.404$, $0.299 \rightarrow 0.296$), and the common tier does not move at all ($0.158 \rightarrow 0.158$). A single scalar cannot repair LocalRetro’s sharp, tier-concentrated collapse, because it applies the same correction everywhere regardless of tier. For MHNreact, the improvement is more evenly spread across tiers, including a real (if partial) gain in the rare tiers ($0.128 \rightarrow 0.101$, $0.117 \rightarrow 0.057$), consistent with MHNreact’s own uniformly-miscalibrated signature (Figure 1) rather than a sharp collapse: a global correction works better when the underlying miscalibration is itself more global. Either way, neither model’s rare tail is fixed by temperature scaling to the degree the overall number suggests, which is exactly the failure mode Mondrian conformal (Section 8) is designed around: a per-tier problem needs a per-tier fix, not a global one.

Tier	n	ECE (raw)	ECE (T-scaled)
<i>LocalRetro</i> (fitted $T = 1.513$)			
Very rare (1–5)	46	0.408	0.404
Rare (6–20)	84	0.299	0.296
Common (21–100)	243	0.158	0.158
Very common (100+)	2,131	0.040	0.031
Overall	2,504	0.047	0.028
<i>MHNreact</i> (fitted $T = 1.744$)			
Very rare (1–5)	917	0.128	0.101
Rare (6–20)	403	0.117	0.057
Common (21–100)	548	0.104	0.034
Very common (100+)	636	0.060	0.037
Overall	2,504	0.099	0.035

Table 3: Temperature scaling, evaluated per tier on a held-out calibration/test split (independent of the main-text tables’ binning scheme, so these ECE values are not directly comparable to Tables 1/2 in absolute terms; the comparison that matters is raw vs. T-scaled within this table). Both fitted temperatures exceed 1, consistent with both models being overconfident on average.

Single-architecture detail (LocalRetro’s and MHNreact’s own absolute-count-tier calibration gap, AUROC-by-tier, and fine-grained 10-quantile-bin figures, plus the full absolute-count tier table) is in Supplementary Information Section 10.3, since the rank-based comparison above is the primary, confound-free cross-model result.

6 True Distribution Shift: A Held-Out-Reaction-Class Experiment

Section 5 uses template training frequency as a continuous proxy for distributional novelty, an honest but indirect measure: a template seen once or twice is treated as functionally out-of-distribution without ever being genuinely absent from training. This section reports a stronger test: reaction classes held out entirely from training, so the templates involved are either structurally absent from the model’s vocabulary (LocalRetro) or present but never trained on (MHNreact, see below).

Design

Reaction classes follow the standard Schneider et al.¹⁵ 10-class taxonomy, the de facto labeling used across the USPTO-50K literature. Two classes were selected by a data-driven criterion, not chemical interest: template exclusivity. For each candidate class, we checked what fraction of its reactions use templates found *nowhere else* in the dataset, in both models’ own template vocabularies independently (LocalRetro’s 671-template space and MHNreact’s 11,800-template space are extracted separately and need not agree). Class 9, the largest candidate, turned out to be only 67.3% exclusive in LocalRetro’s vocabulary (its templates are shared with other classes over 30% of the time) and was rejected as a contaminated test. Class 4 (Heterocycle formation, n=909) and Class 8 (Oxidations, n=811 literature / 811 found) came back 99.8–100% exclusive in *both* models’ vocabularies independently, and were selected for that reason.

For each held-out class, all of its reactions (pooled across the original train/valid/test splits) are removed from training entirely and used as a genuine out-of-distribution (OOD) test set. A second, in-distribution (indist) test set, drawn from the original test split minus the held-out class, uses the same retrained model to confirm the retrain did not simply degrade the model overall.

The two architectures required different implementations because their template vocabularies are constructed differently, a distinction that matters for the results below rather than a methodological inconsistency. **LocalRetro** extracts its template vocabulary fresh from whatever training data it is given, so excluding a class’s reactions from training removes that class’s exclusive templates from the vocabulary entirely: a held-out template is structurally impossible to predict, not merely unlikely. **MHNreact**’s template vocabulary (11,800 templates) is fixed once, upfront, from the full dataset before any training run; excluding a class’s reactions from training leaves its templates present as valid output classes with random, untrained embeddings: a held-out template is astronomically unlikely to be predicted correctly, but not architecturally forbidden.

Results

Model	Class	Test	n	Accuracy	AUROC	ECE
LocalRetro	4 (Heterocycle formation)	OOD	909	0.00% (0/909)	undefined (0 correct)	0.376
LocalRetro	4 (Heterocycle formation)	Indist	4,916	53.30% (2,620/4,916)	0.726 [0.712, 0.740]	0.040
LocalRetro	8 (Oxidations)	OOD	811	0.00% (0/811)	undefined (0 correct)	0.390
LocalRetro	8 (Oxidations)	Indist	4,925	52.89% (2,605/4,925)	0.718 [0.703, 0.731]	0.044
MHNreact	4 (Heterocycle formation)	OOD	909	0.11% (1/909)	not interpretable ($n_{\text{correct}} = 1$)	0.268
MHNreact	4 (Heterocycle formation)	Indist	4,916	47.29%	0.730 [0.716, 0.744]	0.097
MHNreact	8 (Oxidations)	OOD	811	0.25% (2/811)	not interpretable ($n_{\text{correct}} = 2$)	0.257
MHNreact	8 (Oxidations)	Indist	4,925	46.50%	0.722 [0.707, 0.736]	0.095

Table 4: Held-out-reaction-class results. “OOD” = the held-out class’s reactions, entirely absent from training. “Indist” = the original test set minus the held-out class, evaluated on the same retrained model, confirming the retrain did not degrade the model generally. AUROC and ECE are computed exactly as in Sections 4–5. LocalRetro’s OOD AUROC is mathematically undefined (zero correct predictions in either class, no positive class to rank). MHNreact’s OOD AUROC is omitted for the same underlying reason it would be misleading to report: with only 1 and 2 correct predictions respectively, an AUROC value (and any confidence interval around it) reflects the rank of a single point, not a stable statistic, even though the computation runs and returns a plausible-looking number. The OOD story in this table rests on accuracy and ECE, both well-supported at these sample sizes.

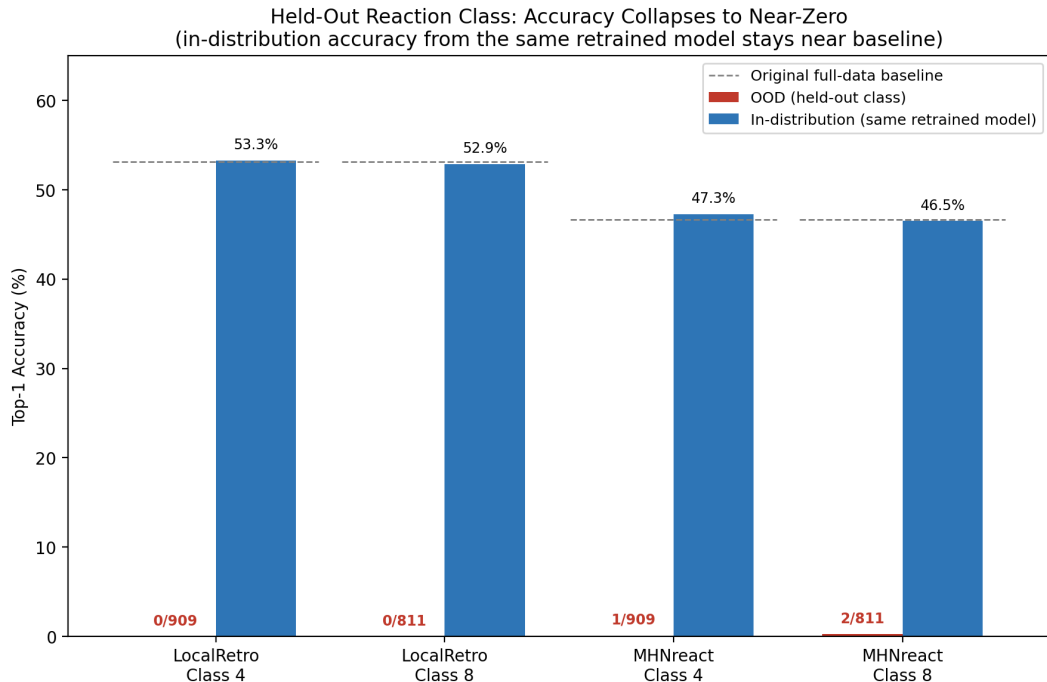


Figure 2: Held-out-class accuracy, all four model/class cells. In-distribution accuracy (blue) tracks each model’s original full-data baseline (dashed line) closely, confirming a healthy retrain. OOD accuracy (red) collapses to 0/909 and 0/811 for LocalRetro, and 1/909 and 2/811 for MHNreact, far below the frequency proxy’s very-rare-tier accuracy (10.0% LocalRetro, 28.4% MHNreact, not shown), confirming genuine distribution shift is a strictly harder regime than the proxy approximated.

In-distribution accuracy confirms the retrain is healthy, not just broken from less data. All three completed indist runs land within a point or two of their model’s original full-data baseline (LocalRetro: 53.30% vs. 53.11% original; MHNreact Class 4: 47.29% vs. 46.69%; MHNreact Class 8: 46.50% vs. 46.69%). Whatever happens on the held-out class is attributable to that class specifically, not to a generally degraded model.

On the held-out class itself, accuracy collapses far more completely than the frequency proxy predicted, and replicates exactly across two independent classes. The frequency proxy’s very-rare tier (freq 1–5) showed 10.0% accuracy for LocalRetro and 28.4% for MHNreact, already a severe collapse, but not remotely as severe as what genuine novelty produces: LocalRetro gets **zero out of 909 (Class 4)** and **zero out of 811 (Class 8)** correct, and MHNreact gets 1/909 and 2/811. LocalRetro’s 0% is not a one-class artifact: it reproduces exactly across two chemically unrelated held-out classes (heterocycle formation and oxidation reactions), which is what the architectural explanation below predicts and a single-class result could not have confirmed on its own. Genuine distribution shift is a strictly harder regime than the frequency proxy approximated, not just a noisier version of the same thing.

The architectural difference in vocabulary construction predicts the exact difference in outcome. LocalRetro’s 0/909 and 0/811 are not approximations of "very unlikely": they are architecturally guaranteed, since the correct templates do not exist as candidate outputs at all, and this holds identically across both held-out classes. MHNreact’s 1 and 2 correct predictions (out of 909 and 811, respectively) are the residue of astronomically unlikely but not-forbidden guesses against randomly-initialized template embeddings. This is a clean, mechanistic explanation that falls directly out of the two models’ template-vocabulary designs, not a post-hoc story fit to the numbers, and the replication across two chemically unrelated classes for LocalRetro rules out a one-class coincidence.

Calibration is severely broken in the direction that matters most for deployment. LocalRetro’s OOD ECE is 0.376 (Class 4) and 0.390 (Class 8): the model reports substantial confidence (mean confidence roughly 38–39%) on classes where it is correct literally 0% of the time, consistently across both.

MHNreact’s OOD ECE (0.268, 0.257) is similarly severe, if slightly less extreme. Both models are confidently wrong under genuine novelty, which is the practical failure mode a deployed synthesis-planning tool cannot afford. The confidence distribution underlying these ECE values is shown in Supplementary Information Section 10.5.

7 Perturbation UQ Does Not Rescue the Rare Tail (Both Models)

The calibration collapse in Section 5 motivates an obvious question: can perturbation-based uncertainty quantification, MC dropout or a deep ensemble, recover discrimination in the rare tail where plain softmax confidence fails? The answer is no for both models, and for MHNreact, both perturbation signals do something worse than fail: they invert.

MC Dropout

For LocalRetro, no MC dropout derived signal outperformed softmax confidence. Bootstrap CIs confirm the baseline vs. MC-average comparison (0.7104 vs. 0.7063) is not a real difference: the intervals overlap completely. The very-rare tier showed an elevated MC dropout variance AUROC (0.729), but this did not survive Bonferroni correction (corrected CI [0.493, 0.901], just spanning 0.5) and was subsequently refuted by the deep ensemble below.

MC dropout was extended to MHNreact by reactivating its `nn.Dropout` submodules at inference (Section 2.3), verified against the model’s actual source rather than assumed. Per-signal AUROC for both models is shown in Supplementary Information Section 10.6.

MC dropout fails to improve on baseline for both models, but the character of that failure differs. For LocalRetro, all derived signals are weak-but-harmless: they simply carry less information than plain softmax confidence. For MHNreact, the failure is architecturally sharper: MC-averaging measurably *degrades* confidence (a real, statistically significant degradation, not just a null result: paired DeLong test diff = -0.0136 , $p = 7.1 \times 10^{-11}$), and dropout variance is not uninformative but actively backwards: higher disagreement across stochastic passes predicts *correct* answers more often than incorrect ones, both overall (AUROC 0.330) and in the very-rare tier (AUROC 0.361, Bonferroni CI [0.328, 0.396], entirely below 0.5).

This inversion is boundary-robust, not an artifact of one tier definition: sweeping the very-rare cutoff from $\text{freq} \leq 3$ to $\text{freq} \leq 20$, AUROC stays between 0.329 and 0.372 and the Bonferroni-corrected CI excludes 0.5 at every cut tested (Figure 3, right panel). The effect size is modest, roughly 0.13–0.17 below random, not a strong signal in the opposite direction, but it is a stable, not a fragile, finding. One benign alternative mechanism worth naming: MHNreact’s very-rare tier likely contains molecules with several near-tied candidate templates (any of which counts as correct), and dropout passes may disagree specifically on those multi-valid-template cases, which are also more likely to be scored correct regardless of which candidate is sampled. That would produce an apparent inversion without implying the model’s epistemic uncertainty is architecturally backwards; the data cannot yet distinguish this from a genuine inversion.

Deep Ensemble

To test whether epistemic uncertainty from model variance recovers discrimination in the rare-template regime where MC dropout failed, we trained 4 independent seeds of each model (LocalRetro seeds 2–5; MHNreact seeds 1–4, same architecture, hyperparameters, and verified early-stopping protocol as the baseline, differing only in initialization).

For LocalRetro, averaging predictions across the 4 seeds improves overall AUROC from 0.710 to 0.727, a real gain confirmed by a paired DeLong test and paired bootstrap (diff = 0.017, 95% CI [0.010, 0.023], $p = 3.6 \times 10^{-8}$): the one perturbation-based approach that genuinely helps, though the gain is modest. Ensemble disagreement does not recover signal in the very-rare tier (AUROC 0.601, Bonferroni CI [0.361, 0.812], spans 0.5), closing the loop on the MC dropout variance hypothesis above: disagreement across independently trained models carries no more signal in the rare tier than disagreement across stochastic dropout passes.

For MHNreact, the identical test finds no ensemble-mean improvement at all (diff = 0.001, 95% CI $[-0.006, 0.007]$, $p = 0.868$). Worse, MHNreact’s ensemble disagreement is not a clean null in the very-rare tier as it is for LocalRetro; it is inverted (AUROC = 0.427, Bonferroni CI [0.394, 0.461], entirely below 0.5), the same direction of surprise as the MC dropout variance inversion above. Very-rare-tier AUROC with Bonferroni CIs for both models is shown in Supplementary Information Section 10.6.

A second, completely independent 4-seed MHNreact ensemble (seeds 5–8, trained separately from the first) was run to test whether this inversion is a property of the model or an artifact of one particular set of

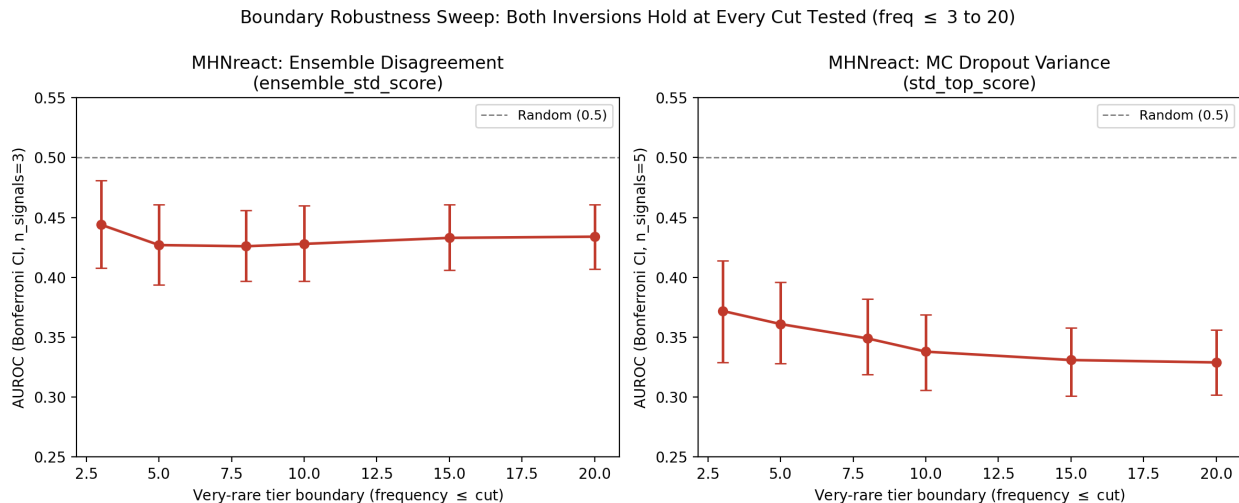


Figure 3: Boundary robustness sweep for both MHNreact inversion claims: AUROC and Bonferroni-corrected CI recomputed at six different very-rare-tier boundary definitions (freq $\leq 3, 5, 8, 10, 15, 20$). Both inversions (MC dropout variance, right; ensemble disagreement, left) hold at every cut tested: the CI excludes 0.5 in all six cases for both signals. A claim that holds only at one arbitrary boundary would be fragile; this one does not.

random initializations. It replicates closely (Supplementary Information Table 12), and a boundary sweep on this second ensemble also replicates: the inversion is not a one-ensemble or one-cutoff artifact.

Synthesis: A New Cross-Architecture Divergence

Deep ensembling was expected to behave similarly across two architecturally distinct models, mirroring the shared OOD calibration-degradation *direction* established in Section 5. It does not. LocalRetro’s ensemble mean provides a small, real overall improvement, and its ensemble disagreement is a clean null in the rare tier. MHNreact’s ensemble mean provides no improvement at all, and its ensemble disagreement is backwards in its very-rare tier: a modest effect (0.073 below random, not a strong reversal) but a statistically robust one, confirmed under a boundary sweep *and* a completely independent second ensemble.

One caveat remains regardless of how many seeds or boundaries confirm the pattern: a benign alternative mechanism is possible, structural rather than stochastic (the multi-valid-template near-ties described above), and so not something re-training or re-cutting the tier would rule out. What the evidence supports, stated at the strength it earns: an inversion that has replicated across tier-boundary choices *and* two independently trained ensembles for the deep-ensemble signal, and across every boundary tested for the MC dropout signal, arguing against assuming perturbation-based UQ fixes transfer across architectures without re-validation, though the underlying mechanism remains open. Both models still agree on the paper’s central finding, confidence calibration degrades in the rare-template tail, but the mechanisms available to detect or correct that degradation via perturbation are architecture-specific, not a shared property of template-based retrosynthesis models.

Detailed per-signal comparison tables, ROC curves, distribution plots, and the overall (non-tier) ensemble AUROC figure are in Supplementary Information Section 10.6.

8 What Works: Mondrian Conformal Selective Prediction (Both Models)

Section 7 showed that the obvious fixes, MC dropout and deep ensembling, do not rescue confidence in the rare tail, and can actively mislead in MHNreact. This section shows what does work: abstaining where no threshold can deliver a reliable guarantee, rather than reporting a false one.

Risk-Coverage Motivation

A single fixed target (below) is a snapshot; the risk-coverage curve shows the full tradeoff between coverage (fraction of predictions kept) and risk (error rate among kept predictions) as the confidence threshold sweeps from strictest to most permissive, and motivates why a per-tier approach is necessary at all. At a fixed 20% risk target, LocalRetro’s overall (non-tiered) curve supports keeping 22.6% of predictions; MHNreact’s supports only 14.8%, despite MHNreact having the *higher* overall baseline AUROC (0.738 vs. 0.710): AUROC is threshold-agnostic, while the risk-coverage curve also reflects the base error rate (MHNreact 53.3% vs. LocalRetro 46.9%). Per-tier, LocalRetro’s very-rare-tier curve never drops to the 20% target line even at very low coverage, visual confirmation of why Mondrian correctly abstains on that tier rather than reporting a false guarantee (full curves in Supplementary Information Section 10.8).

Global vs. Mondrian Thresholds

We test whether frequency-stratified (Mondrian) conformal selective prediction can restore per-tier error guarantees, using a combined softmax + MC-dropout-variance nonconformity score fit on a held-out calibration half-split, against a 20% target error rate. For LocalRetro, a global (marginal) threshold catastrophically fails: it keeps 9% of very-rare-tier predictions with a **100% error rate**. The Mondrian approach correctly abstains entirely on the very-rare tier (where no threshold can achieve the target given the base accuracy of 10%), and holds the error near target in the common and very-common tiers.

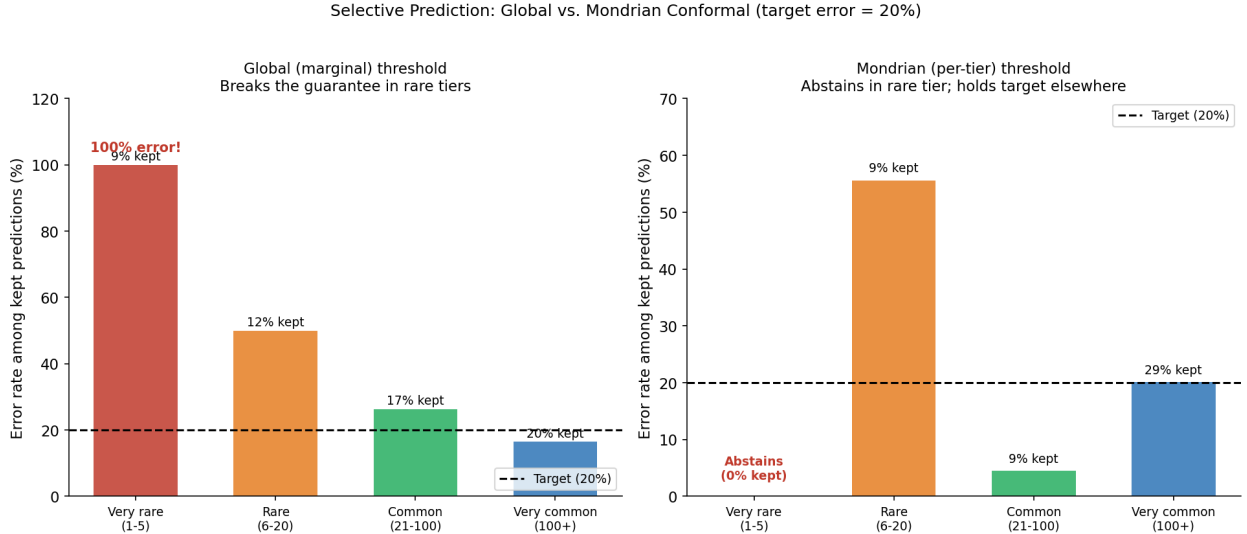


Figure 4: LocalRetro selective prediction under a 20% target error rate. *Left*: A single global threshold keeps 9% of very-rare predictions with a 100% error rate, far above the target. *Right*: Mondrian (per-tier) thresholds correctly abstain on the very-rare tier and hold near the target in other tiers. Numbers above bars show the fraction of predictions kept (coverage).

The identical harness run on MHNreact replicates the qualitative story: the global threshold badly overshoots the target in the very-rare tier (63.4% error kept vs. 20% target), and Mondrian correctly recognizes

this tier is hopeless and abstains entirely. MHNreact’s global-vs-Mondrian figure and its naive per-tier results table are in Supplementary Information Section 10.7.

Honesty Check: Hoeffding-UCB Correction

The quantitative story is not a clean win on this single calibration/test split: Mondrian’s rare tier (30.6% error kept) and common tier (22.0%) both land *above* the 20% target. Repeating the split 20 times and averaging shows the common-tier miss was single-split noise (resolves to 18.6% on average, within target), but the rare tier’s miss is real and repeats (29.0% averaged error). Neither a stricter nominal target nor dropping the inverted `std_top_score` feature fixed this; tightening the target actually collapsed coverage toward zero while the held-out error of the few remaining predictions got *worse*. This pattern indicates a sample-size/signal-strength limitation, not a mistuned parameter: the rare tier has roughly 390 calibration molecules and a 51% base error rate, close enough to chance that the logistic combiner cannot reliably find a threshold that generalizes from calibration to test.

This kind of finite-sample slack is a documented property of this exact conformal method, not a symptom specific to MHNreact. We tested the standard correction directly: instead of selecting the most permissive threshold whose *empirical* calibration error stays under target (optimistic, since searching over many candidate thresholds and keeping the best-looking one has a selection effect), we selected the most permissive threshold whose Hoeffding *upper confidence bound* on the error rate stays under target, at 90% confidence ($\delta = 0.1$).

Tier	Naive: coverage / true error	UCB-corrected: coverage / true error
Very rare (1–5)	0.0% (abstains)	0.0% (abstains)
Rare (6–20)	8.3% / 29.0% (above target)	0.0% (abstains)
Common (21–100)	26.8% / 18.6%	0.0% (abstains)
Very common (100+)	58.7% / 19.5%	32.8% / 12.2%

Table 5: Naive vs. Hoeffding-UCB-corrected Mondrian thresholds, averaged over 20 random calibration/test splits. The corrected threshold respects the 20% target everywhere it does not abstain, at the cost of abstaining on both the rare and common tiers entirely rather than keeping a modest fraction with an unreliable guarantee.

This makes the diagnosis precise rather than qualitative: the naive threshold was not simply “a bit optimistic” in the rare and common tiers; it was trading an invalid guarantee for coverage that a valid procedure cannot support at all with the available calibration data. Only the very-common tier (largest calibration sample, $n \approx 645$ per split) supports a real, valid selective-prediction filter.

Two further candidate fixes were tested against the same repeated-split diagnostic, and both were ruled out rather than left untested. **Calibration/test split ratio:** moving from 50/50 to 60/40, 70/30, and 80/20 (giving the rare tier’s threshold search more calibration data at the cost of a smaller held-out test set) did not help; the true held-out error *rose* monotonically with calibration fraction (29.0% $\rightarrow$ 30.2% $\rightarrow$ 31.0% $\rightarrow$ 33.9%), since shrinking the test set only adds noise to the honesty check without adding real signal to the underlying problem. **K-fold cross-conformal** ($K = 5$ and $K = 10$, where every molecule is calibration data in most folds and test data in exactly one) performed *worse*, not better: true held-out error rose to 36.3% ($K = 5$) and 43.5% ($K = 10$), and the previously-fine common tier was pushed to the edge of the target (20.1–20.3%) as well. Cross-conformal usually helps by enlarging the calibration set per fold, but here that benefit is outweighed by running the same optimistic threshold-search procedure multiple times against smaller, noisier held-out folds; with a base error rate this close to chance (51%), the repetition compounds noise rather than averaging it away.

Two further approaches were tested, differing in kind from the four above: rather than resampling the same information, they attempt to add genuinely new information or restructure the tiers themselves. **A richer nonconformity combiner** (adding `agreement_rate`, `predictive_entropy`, and `ensemble_std` to the existing `uq + std_top_score` features) barely moved the needle: true held-out error went from 29.0% to 28.3%, still far from the 20% target. **Merging the very-rare and rare tiers into one larger tier**

($n \approx 2,620$) backfired outright: rather than diluting the difficulty with more data, it abstained on every one of the 20 test splits, since the merge also blends in the very-rare tier’s 70.8% base error rate, making the target harder to hit with the combined pool, not easier.

Six independent attempts (stricter nominal target, dropping the inverted `std_top_score` feature, calibration/test split ratio, K-fold cross-conformal, a richer feature set, and tier merging) now converge on the same conclusion: this is not a fixable methodology problem within the current data. The rare tier’s discriminative signal is too weak at $n \approx 785$ total for any standard resampling or restructuring scheme to produce a threshold that reliably generalizes at a 20% target. The practical implication for deployment: the naive Mondrian threshold should not be used as-is for MHNreact’s rare or common tiers; either the Hoeffding-UCB-corrected threshold should be used as-is (honest, but close to abstaining in those tiers), or substantially more calibration data is needed, most plausibly from a larger benchmark than USPTO-50K’s fixed 5,007-reaction test set, since no resampling or restructuring of the existing pool closes the gap.

9 Discussion

Several results (the perturbation-UQ negative result, the calibration collapse shared across two architectures, and the divergent perturbation-UQ behavior discovered along the way) combine into a coherent picture of template-based retrosynthesis confidence.

- Softmax confidence is a real discriminator overall in both models: LocalRetro AUROC 0.710 [0.696, 0.724], MHNreact AUROC 0.738 [0.724, 0.752]. Both are moderate but statistically robust signals.
- Both models show degradation concentrated in the rare template tail. Using the rank-based vocabulary-matched tiers (Table 2), the debiased-ECE degradation ratio (rare/high) is $3.2\times$ for LocalRetro and $2.9\times$ for MHNreact (similar by that denominator), but the rare/mid ratio ($7.3\times$ vs. $1.8\times$) shows the underlying failure shapes differ substantially. The shared degradation *direction*, not a single ratio, is the cross-architecture finding.
- The two models exhibit different failure signatures within that shared trend (Figure 1): LocalRetro is well-calibrated in-distribution and collapses under novelty; MHNreact is mildly but uniformly miscalibrated across all tiers, with no sharp collapse. These are two architecturally distinct miscalibration regimes sharing one qualitative property: worst calibration in the rare tail.
- MC dropout adds no discrimination over plain softmax confidence overall for LocalRetro, and the $20\times$ inference cost is not justified for in-distribution error detection. For MHNreact, MC dropout is not merely useless but actively counterproductive: MC-averaging degrades confidence, and dropout variance is inverted.
- A 4-seed deep ensemble provides a real overall improvement for LocalRetro ($+0.017$ AUROC, $p = 3.6 \times 10^{-8}$), the one perturbation-based approach that genuinely helps, though the gain is modest. This does not generalize to MHNreact: no ensemble-mean improvement, and an ensemble-disagreement signal that is inverted rather than merely null in the very-rare tier, replicated across boundaries and a second independent ensemble. Deep ensembling is not a portable UQ fix across architectures, even though both models share the same underlying calibration-degradation direction.
- Mondrian (per-tier) conformal selective prediction is the practical, architecture-agnostic fix. A global threshold catastrophically over-keeps very-rare predictions (100% error rate on the 9% it keeps for LocalRetro). Per-tier thresholds correctly abstain on the very-rare tier and hold near the error target elsewhere for both models, though the Hoeffding-UCB check shows MHNreact’s naive per-tier guarantee is itself optimistic in the rare and common tiers, a second-order but genuine caveat.

Template-based retrosynthesis models are honest about what they know. That is probably why MC dropout and deep ensemble disagreement, in-distribution, find little room to add beyond softmax confidence in LocalRetro. The OOD calibration degradation is the most consequential result for deployment: confidence becomes poorly calibrated exactly on the rare, novel templates where a synthesis planning tool most needs a reliable “I don’t know” signal. This degradation is shared across two architecturally distinct models with matched ECE ratios of $\sim 3\times$, suggesting it reflects the task’s distributional structure (a template vocabulary with a long-tail frequency distribution creates a hard coverage boundary) rather than any specific modeling choice. The Mondrian conformal fix operationalizes this diagnosis into a practical decision rule with per-tier guarantees applicable to any model in this family. Deep ensembling and MC dropout, however, do not share that portability: they help LocalRetro modestly (ensemble) or not at all (dropout), and actively mislead for MHNreact, so any deployment relying on perturbation-based disagreement as a rare-tier safety signal needs architecture-specific validation rather than an assumption of transfer.

10 Limitations and Next Steps

- All results are on USPTO-50K. Whether the calibration collapse, the perturbation-UQ inversions, and the Mondrian mitigation generalize to a larger, longer-tailed benchmark (USPTO-full, USPTO-MIT) is untested; the failure modes reported here could be more or less severe at scale.

- The held-out-class experiment covers two reaction classes (heterocycle formation, oxidations), selected for template exclusivity rather than chemical breadth. Replication across two chemically unrelated classes for LocalRetro is reassuring but does not establish that every novel reaction type would collapse identically.
- The main-text calibration analysis uses template training frequency as a proxy for distributional novelty. This proxy correlates with but is not identical to genuine distribution shift, and the two should not be treated as interchangeable measures of the same underlying quantity.
- MHNreact’s Mondrian calibration pool is too small in the rare and common tiers ($n \approx 785$) for the naive per-tier threshold to generalize reliably; the Hoeffding-UCB-corrected threshold is honest but abstains on most of the tier as a result. This is a sample-size limitation of the current benchmark, not of the method.
- The mechanism behind MHNreact’s inverted perturbation-based UQ signals (MC dropout variance, ensemble disagreement) is not identified. A benign explanation (near-tied multi-valid-template cases) and a genuine architectural cause remain equally consistent with the data; distinguishing them would require molecule-level inspection rather than further statistical testing.
- The Mondrian nonconformity score is a simple logistic combination of two features (softmax confidence and MC-dropout variance). More expressive combiners were tested only within the current calibration pool’s sample-size constraints and produced negligible improvement; whether a richer score would help given more data is untested.
- Each model’s headline results rest on a single trained checkpoint (with independent ensembles used only for the ensemble-specific analyses). Run-to-run variance beyond what those ensembles capture is not separately quantified.
- Temperature scaling (Section 5) uses a one-vs-rest approximation on the top-1 probability, since the full softmax distribution over the template vocabulary was not retained; results may differ modestly from temperature scaling applied to the full logit vector.

Code and Data Availability

All analysis code (calibration toolkit, AUROC bootstrap, Mondrian conformal harness, quantile binning, paired DeLong testing, temperature scaling, and boundary sweep implementations) will be released at a public repository upon publication: <https://github.com/yamenfargaly-maker/Uncertainty-Quantification-with-Template-Based-Retrosynthesis-Models>. Trained model checkpoints for both architectures (LocalRetro baseline and 4-seed ensemble; MHNreact baseline and two independent 4-seed ensembles) will be released alongside the code. USPTO-50K, the dataset underlying every result in this paper, is publicly available from its original source (Schneider et al. 2016 reaction role classification; Lowe 2012 patent reaction extraction).

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Supporting Information

The following is available free of charge: single-architecture detail, distribution plots, ROC curves, and secondary tables that support the main-text AUROC and ECE numbers without adding new claims (this document, following section). Nothing here contradicts or extends the main text; it exists for readers who want to verify a specific number’s provenance or see a full single-model breakdown.

10.1 S1: Baseline UQ Detail

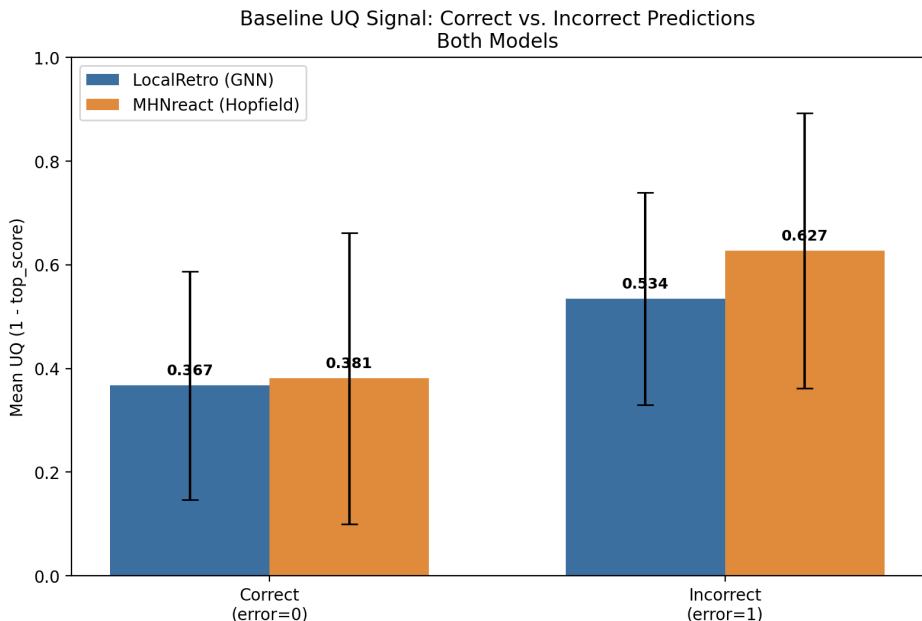


Figure 5: Mean baseline UQ signal ($1 - \text{top_score}$), correct vs. incorrect predictions, both models. Error bars show ± 1 SD. Supports the means reported in Table 1.

10.2 S2: Cross-Model Comparison Detail

10.3 S3: Single-Architecture OOD Calibration Detail

LocalRetro

Frequency tier	n	ECE (EW)	ECE (EM)	Bias floor	ECE (debiased)	MCE
Very rare (1–5)	100	0.392	0.379	0.101	0.291	0.712
Rare (6–20)	176	0.278	0.281	0.077	0.200	0.610
Common (21–100)	486	0.160	0.157	0.047	0.113	0.300
Very common (100+)	4,245	0.038	0.042	0.016	0.024	0.214
Overall	5,007	0.051	0.052	0.014	0.037	0.193

Table 6: LocalRetro: ECE by training-template frequency tier. EW = equal-width bins; EM = equal-mass (quantile) bins; Bias floor = expected ECE under perfect calibration at this n; Debiased = raw EW minus bias floor.

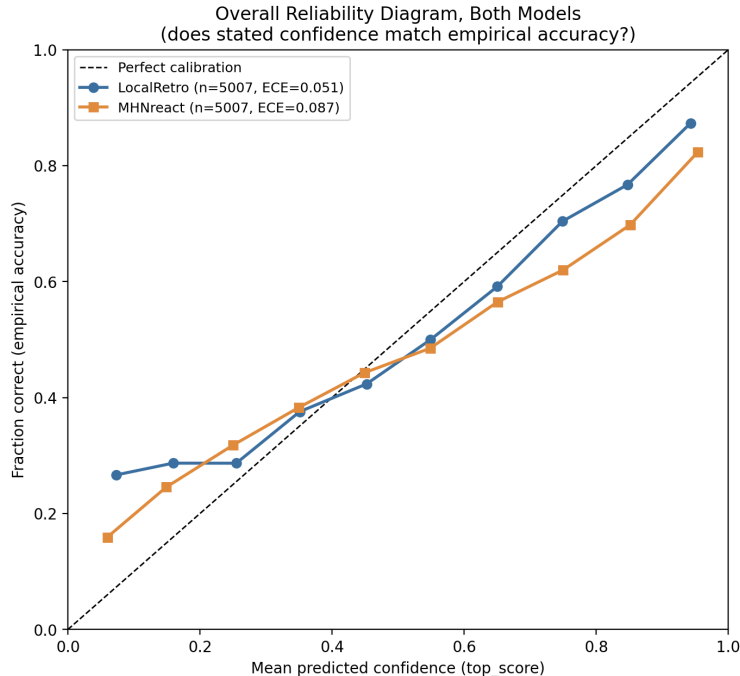


Figure 6: Overall reliability diagram, both models. LocalRetro ($\text{ECE} = 0.051$) is better calibrated on average than MHNreact ($\text{ECE} = 0.087$); both curves sit close to the diagonal in the mid-confidence range and diverge somewhat at the extremes.

MHNreact

Tier	n	Acc	ECE (EW)	ECE (EM)	Bias	ECE (deb)	AUROC
Very rare (1–5)	1,835	0.284	0.129	0.129	0.022	0.107	0.634 [0.608, 0.661]
Rare (6–20)	785	0.454	0.081	0.090	0.036	0.045	0.722 [0.689, 0.757]
Common (21–100)	1,097	0.540	0.097	0.098	0.030	0.068	0.720 [0.688, 0.750]
Very common (100+)	1,290	0.674	0.065	0.068	0.027	0.038	0.741 [0.714, 0.770]
Overall	5,007	0.467	0.087	0.090	0.014	0.073	0.738 [0.724, 0.752]

Table 7: MHNreact calibration by absolute-count frequency tier, retrained with verified early stopping (secondary; see Table 2 for the primary vocabulary-matched comparison). The debiased ECE series ($0.038 \rightarrow 0.068 \rightarrow 0.045 \rightarrow 0.107$ from most to least common) is *not* monotonic on this binning scheme, though the rank-based tiers are cleanly monotonic; this likely reflects tier-composition sensitivity rather than a true non-monotonic calibration trend.

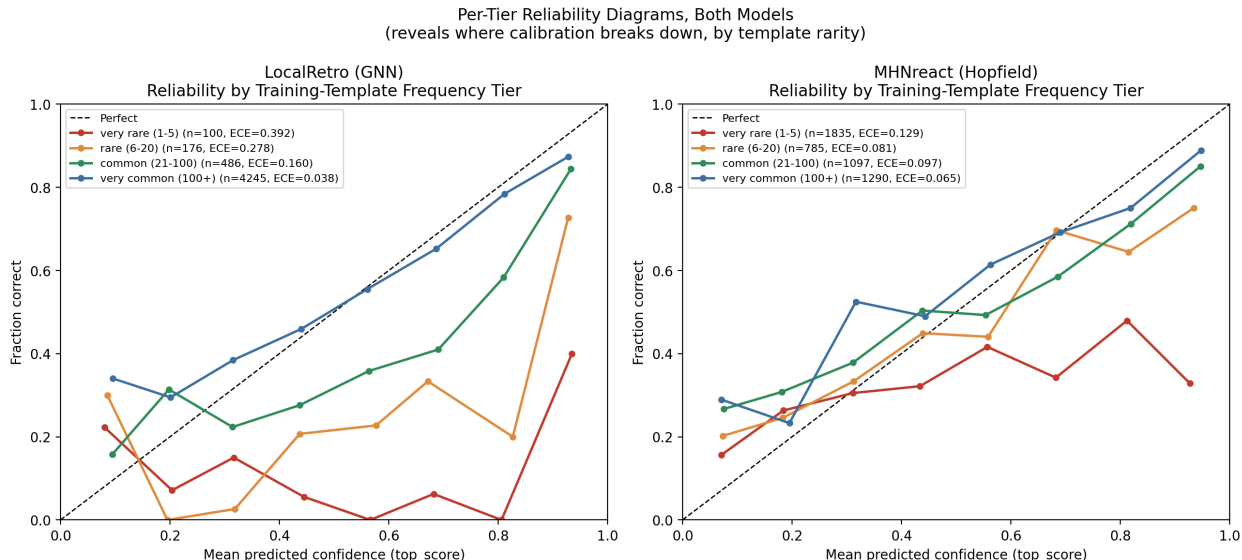


Figure 7: Reliability by training-template frequency tier, both models (side-by-side panels, since each model’s tier boundaries are defined over a different-sized vocabulary: 671 templates for LocalRetro vs. 11,800 for MHNreact, so the tiers are not the same population of reactions and are not meaningfully overlaid on one set of axes). *Left (LocalRetro)*: tiers fan out sharply: very-common (blue) tracks the diagonal closely while very-rare (red, $n=100$) is noisy and far below it, visually matching the twelve-fold debiased ECE spread (0.024 to 0.291). *Right (MHNreact)*: tiers stay comparatively clustered together across the whole confidence range, consistent with the uniformly-mildly-miscalibrated signature.

Signal	AUROC	95% CI	Spearman ρ
1 – top_score (baseline)	0.7104	[0.696, 0.724]	0.3637
1 – mean_top_score (MC avg)	0.7063	[0.692, 0.720]	0.3566
std_top_score	0.5446	[0.530, 0.560]	0.0771
1 – agreement_rate	0.6420	[0.628, 0.657]	0.2698
predictive_entropy	0.5922	[0.576, 0.608]	0.1594

Table 8: LocalRetro: full signal comparison with bootstrap 95% CIs (2,000 resamples).

10.4 S4: Temperature Scaling Detail

10.5 S5: Held-Out-Class Confidence Detail

10.6 S6: Perturbation UQ Detail

MC Dropout

Deep Ensemble

10.7 S7: Mondrian Conformal Detail

10.8 S8: Risk-Coverage Curves

Cross-Architecture Comparison: Rank-Based Vocabulary-Matched Tiers
($n=500/1152/1652/1703$ per tier, same for both models; MHNreact retrained with verified early stopping)

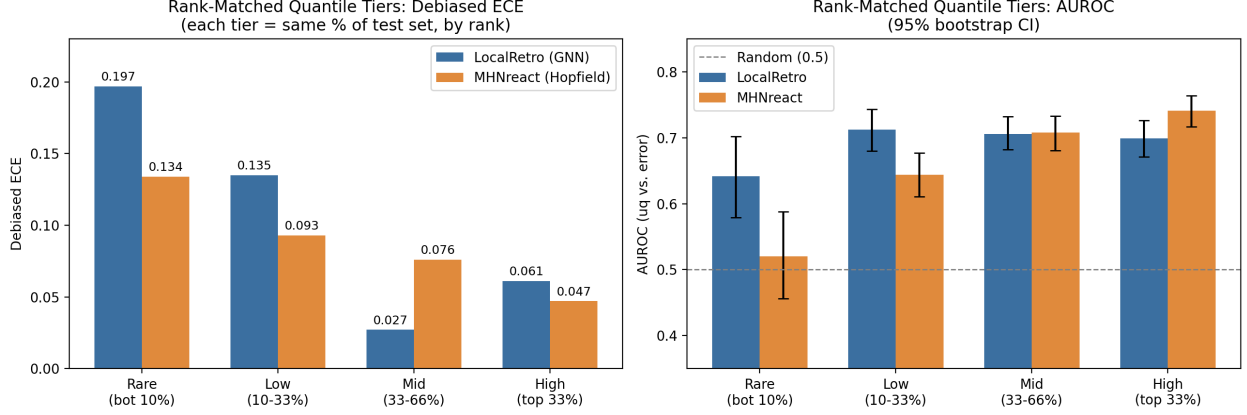


Figure 8: Rank-based vocabulary-matched tier comparison (see main-text Table 2 for the numbers). *Left:* Debiased ECE by tier. Both models have highest ECE in the rare tier, but their profiles differ: LocalRetro shows a sharp drop from rare to mid (0.197 to 0.027), while MHNreact declines smoothly and monotonically (0.134 to 0.047). *Right:* AUROC by tier with 95% bootstrap CIs; both models maintain discrimination above random across all tiers.

Signal	AUROC	95% CI	Spearman ρ
1 – top_score (baseline)	0.7376	[0.724, 0.752]	0.4107
1 – mean_top_score (MC avg)	0.7240	[0.709, 0.738]	0.3872
std_top_score	0.3295	[0.315, 0.345]	−0.2947
1 – agreement_rate	0.7103	[0.697, 0.724]	0.3698
predictive_entropy	0.6942	[0.679, 0.709]	0.3356

Table 9: MHNreact: MC dropout signal comparison, all $p < 10^{-100}$.

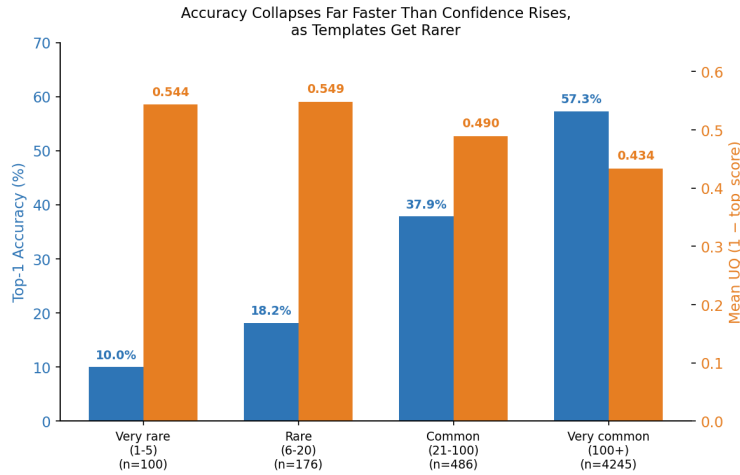


Figure 9: LocalRetro: top 1 accuracy (left axis) and mean UQ (right axis) across four training template frequency tiers. Confidence rises only modestly even as accuracy collapses.

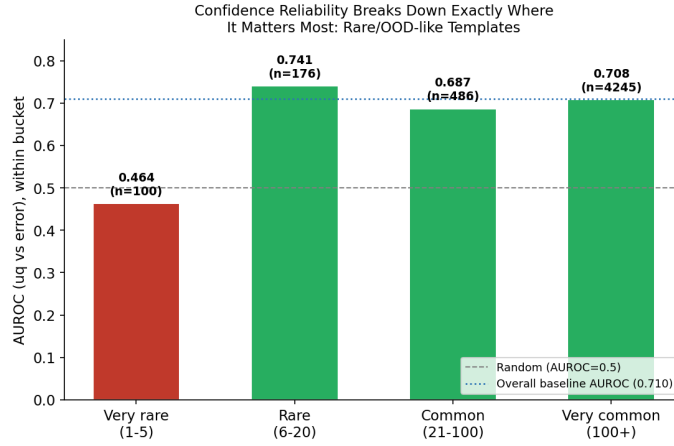


Figure 10: LocalRetro: AUROC (uq vs. error) computed separately within each training frequency tier, with bootstrap 95% CIs.

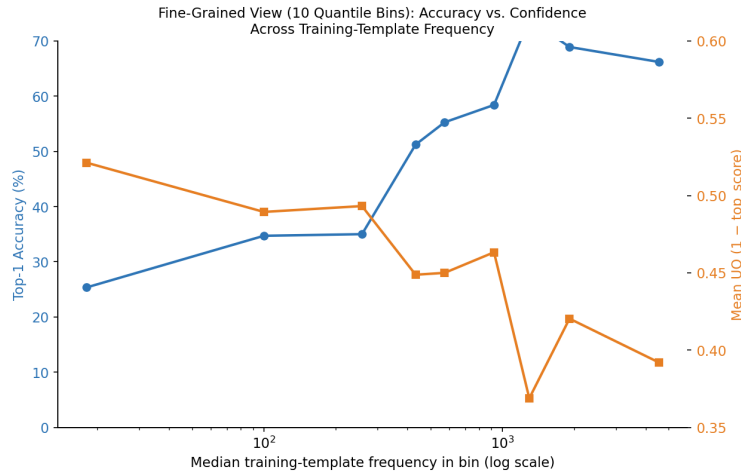


Figure 11: LocalRetro: fine grained (10 quantile bins) accuracy and mean confidence vs. median training template frequency (log x axis).

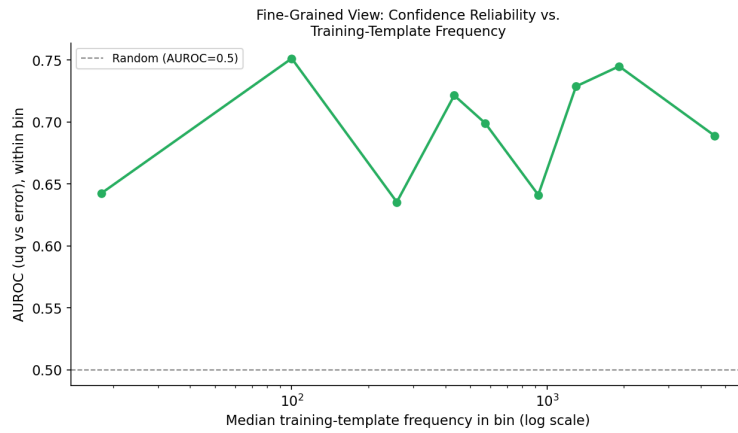


Figure 12: LocalRetro: fine grained within bin AUROC vs. median training template frequency. Reliability stays broadly stable (0.64 to 0.75) outside the most extreme rarity tail.

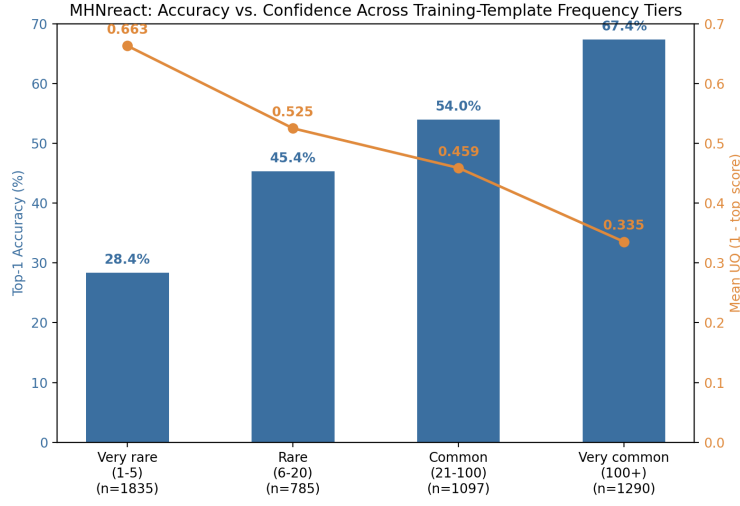


Figure 13: MHNreact: top-1 accuracy (left axis) and mean UQ (right axis) across four training-template frequency tiers. Accuracy rises from 28.4% to 67.4%, a 39-point spread, versus LocalRetro’s 47.3-point collapse from a much smaller very-rare base rate.

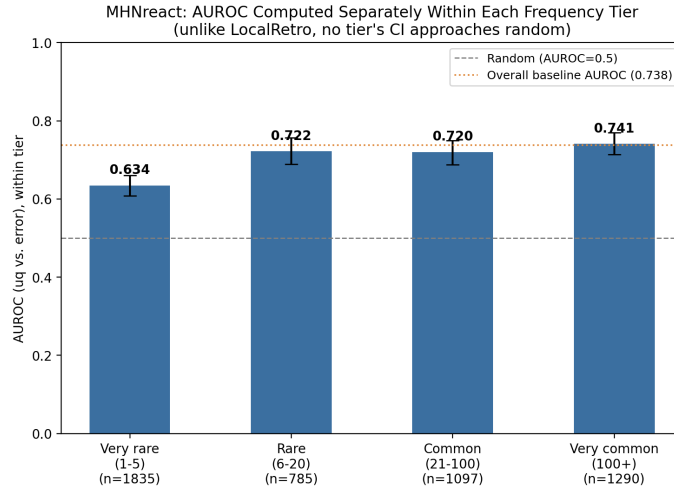


Figure 14: MHNreact: AUROC (uq vs. error) computed separately within each frequency tier, with bootstrap 95% CIs.

Signal	Overall AUROC	Very-rare AUROC	Very-rare Bonf. CI
Baseline ($1 - \text{top_score}$)	0.710 [0.696, 0.725]	0.464	[0.201, 0.761]
Ensemble mean ($1 - \bar{p}$)	0.727 [0.713, 0.741]	0.491	[0.234, 0.766]
Ensemble std (disagreement)	0.549 [0.534, 0.565]	0.601	[0.361, 0.812]

Table 10: LocalRetro deep ensemble results (4 seeds). All very-rare-tier CIs use Bonferroni correction across 3 signals (n=100).

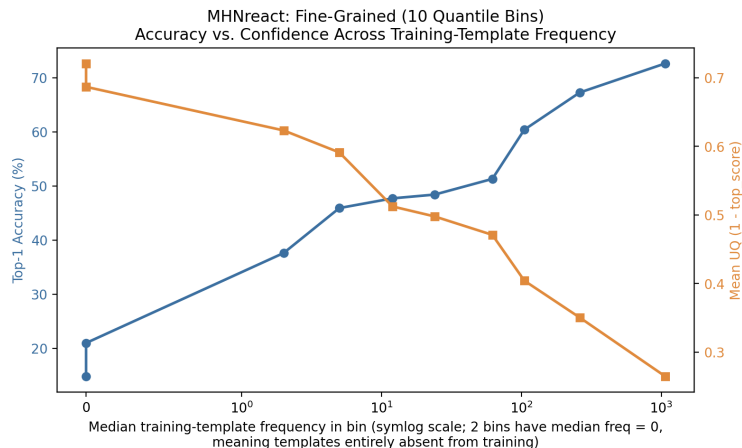


Figure 15: MHNreact: fine-grained (10 quantile bins) accuracy and mean confidence vs. median training-template frequency (symlog x-axis, since 2 bins have median frequency 0).

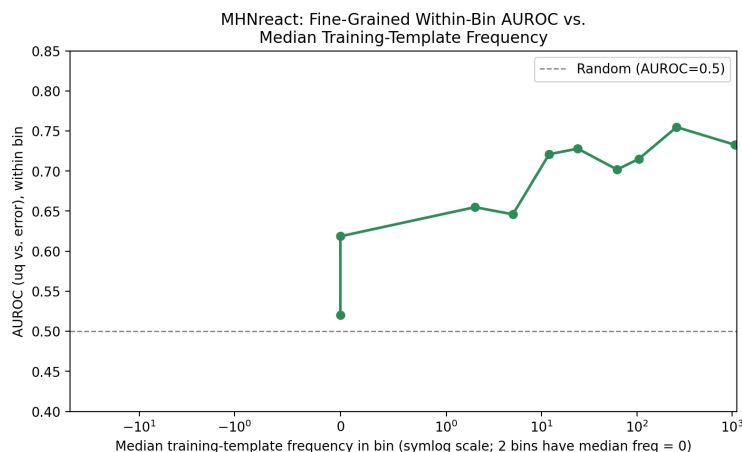


Figure 16: MHNreact: fine-grained within-bin AUROC vs. median training-template frequency. AUROC rises from 0.52 in the zero-frequency bin to a plateau around 0.70–0.75 for established templates, a smoother, more monotonic climb than LocalRetro’s fluctuating 0.64–0.75 range.

Signal	Overall AUROC	Very-rare AUROC	Very-rare Bonf. CI
Baseline ($1 - \text{top_score}$)	0.738 [0.724, 0.752]	0.634	[0.602, 0.665]
Ensemble mean ($1 - \bar{p}$)	0.738 [0.724, 0.752]	0.645	[0.614, 0.676]
Ensemble std (disagreement)	0.499 [0.484, 0.515]	0.427	[0.394, 0.461]

Table 11: MHNreact deep ensemble results (4 seeds: 1–4, seed 0 is the baseline).

Ensemble	Very-rare AUROC	Bonferroni CI
First (seeds 1–4)	0.4270	[0.394, 0.461]
Second (seeds 5–8)	0.4277	[0.394, 0.460]

Table 12: Ensemble-disagreement inversion, two independently trained 4-seed ensembles. Both land within 0.001 AUROC of each other with essentially identical CIs, despite sharing no random initializations.

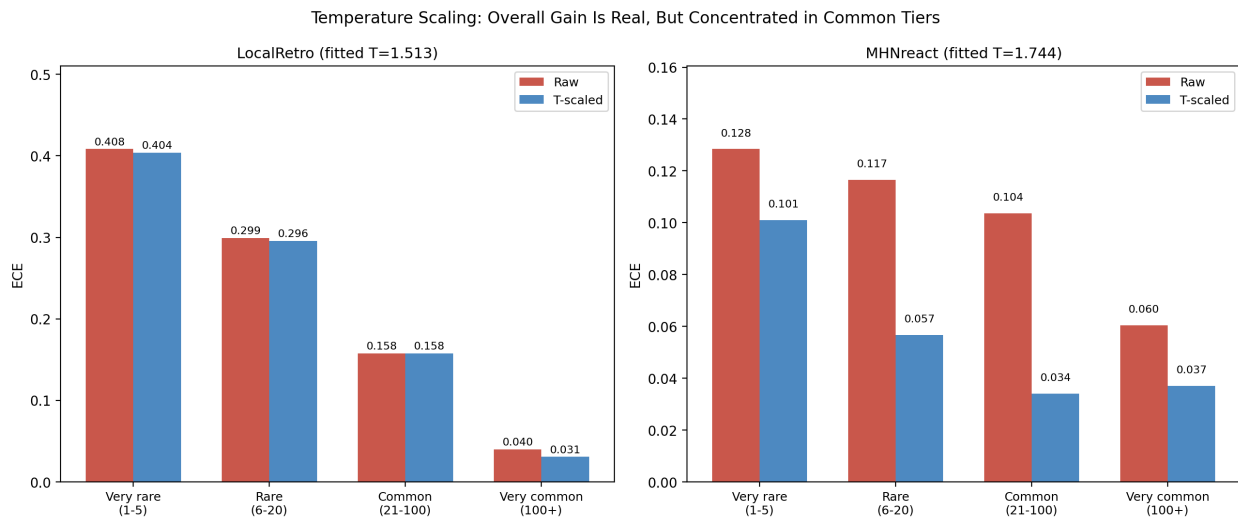


Figure 17: Temperature scaling, raw vs. T-scaled ECE by tier, both models (see main-text Table 3 for the numbers). LocalRetro’s rare tiers barely move and the common tier is unchanged; nearly all of its overall gain comes from the very-common tier. MHNreact improves more evenly across tiers, consistent with its uniformly-miscalibrated signature rather than a sharp collapse.

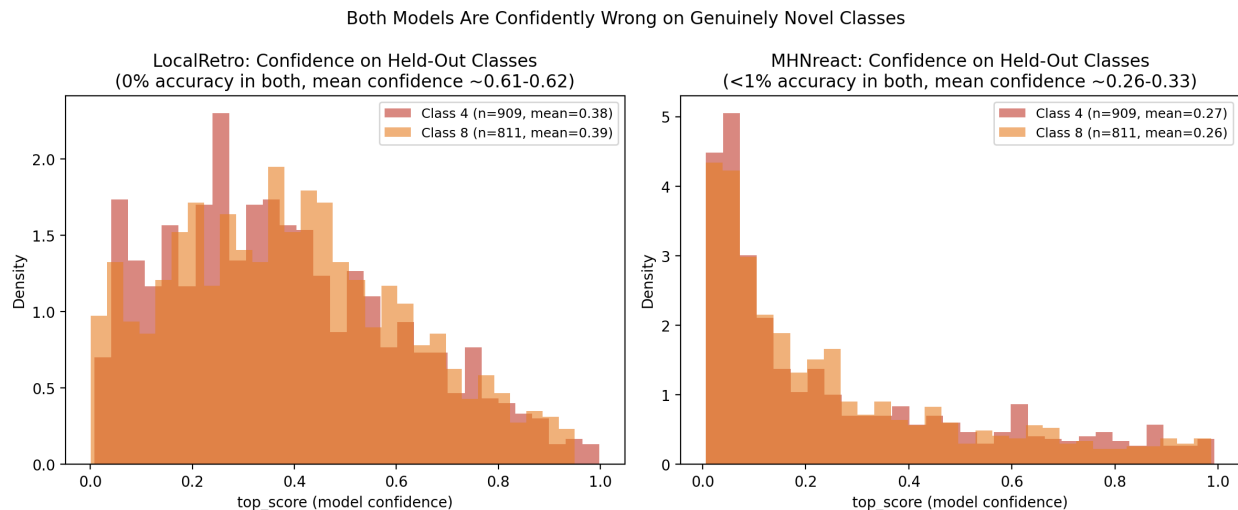


Figure 18: Confidence distribution on the held-out (OOD) test sets, both models (see main-text Table 4 for the ECE values). Despite 0% (LocalRetro) or near-0% (MHNreact) actual accuracy, mean confidence stays well above zero in every case (LocalRetro: 0.376–0.390; MHNreact: 0.259–0.269): both models remain confidently wrong on genuinely novel reaction classes rather than correctly signaling low confidence, the calibration failure mode most relevant to deployment.

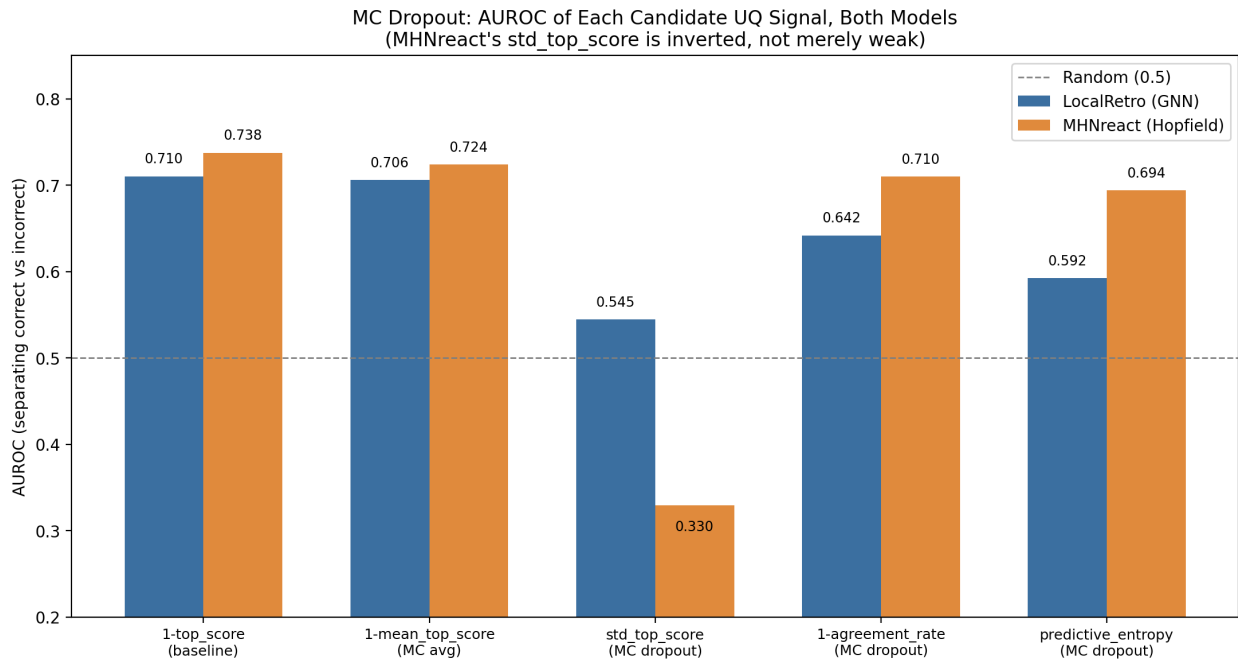


Figure 19: AUROC of each candidate MC dropout signal, both models. MHNreact’s `std_top_score` is not merely weak like LocalRetro’s; it is inverted, falling well below the random-chance line.

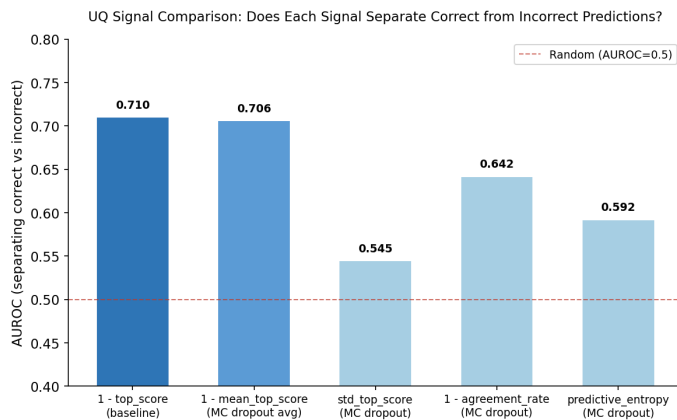


Figure 20: LocalRetro: AUROC of each candidate UQ signal in separating correct from incorrect predictions. Dashed line = random (AUROC = 0.5).

Tier	n (test half)	Global: kept / err	Mondrian: kept / err	Base error rate
Very rare (1–5)	936	4.4% / 63.4%	0.0% (abstains)	70.8%
Rare (6–20)	383	17.5% / 31.3%	22.2% / 30.6%	51.2%
Common (21–100)	543	20.4% / 18.9%	31.9% / 22.0%	45.5%
Very common (100+)	642	31.3% / 13.4%	54.8% / 16.8%	30.2%

Table 13: MHNreact Mondrian conformal results (naive threshold), target error rate 20%. “Global” and “Mondrian” columns show fraction kept (coverage) / error rate among kept predictions. See main-text Table 5 for the Hoeffding-UCB-corrected version of this same analysis.

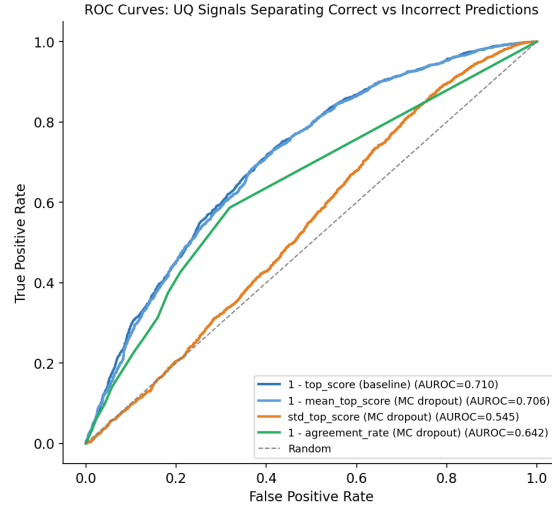


Figure 21: LocalRetro: ROC curves for all evaluated UQ signals.

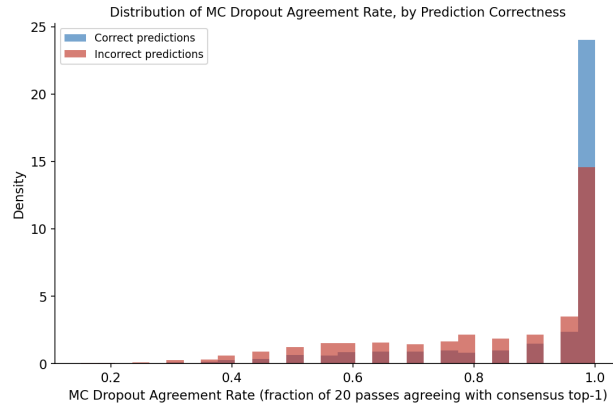


Figure 22: LocalRetro: distribution of MC dropout agreement_rate, by correctness.

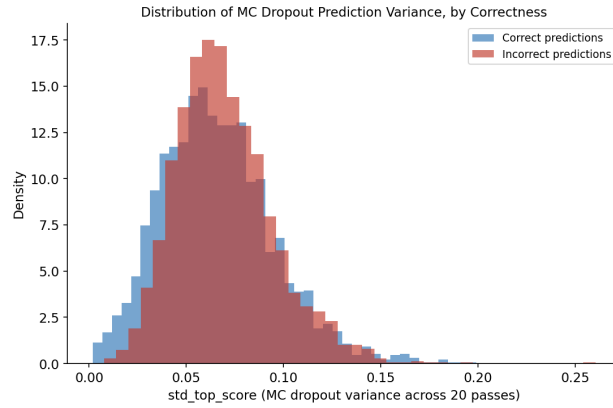


Figure 23: LocalRetro: distribution of MC dropout prediction variance (std_top_score), by correctness.

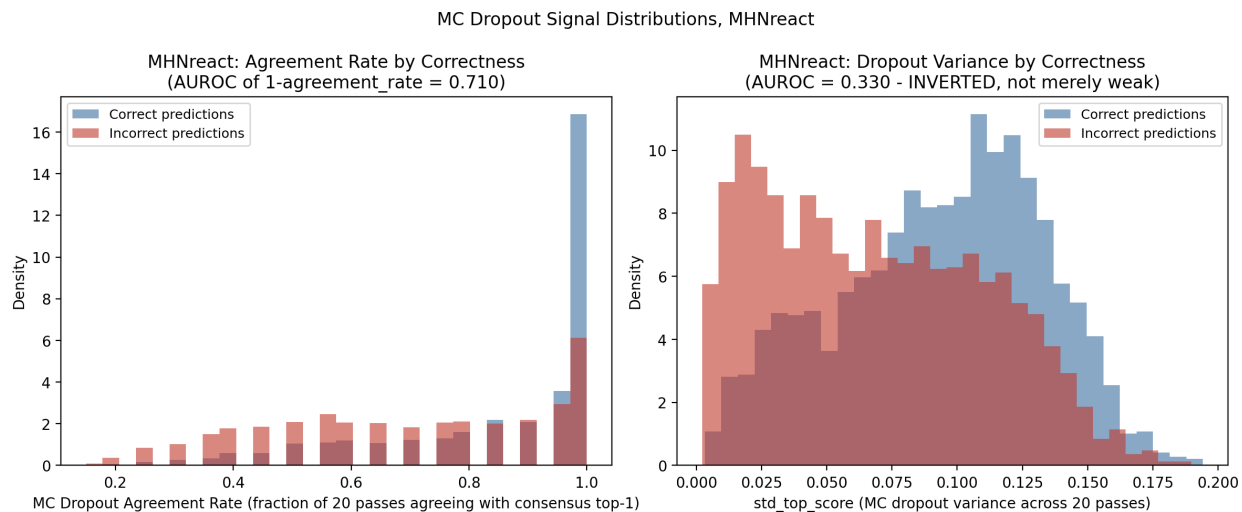


Figure 24: MHNreact: MC dropout signal distributions by correctness. *Left*: agreement rate. *Right*: dropout variance, showing the inversion visually.

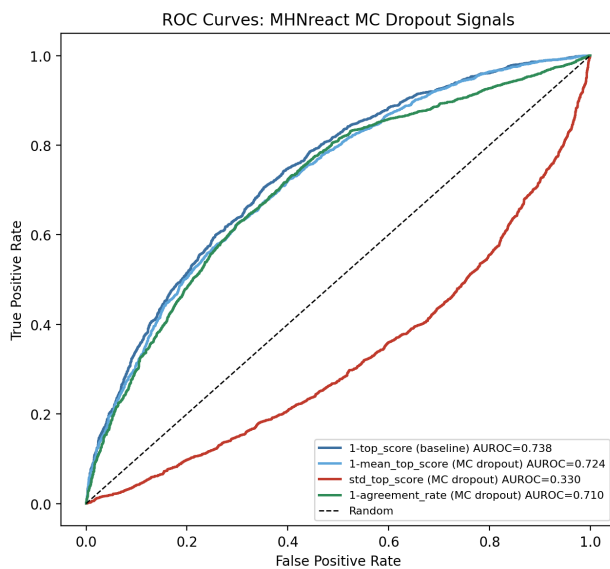


Figure 25: MHNreact: ROC curves for MC dropout signals. The `std_top_score` curve sits visibly below the diagonal.

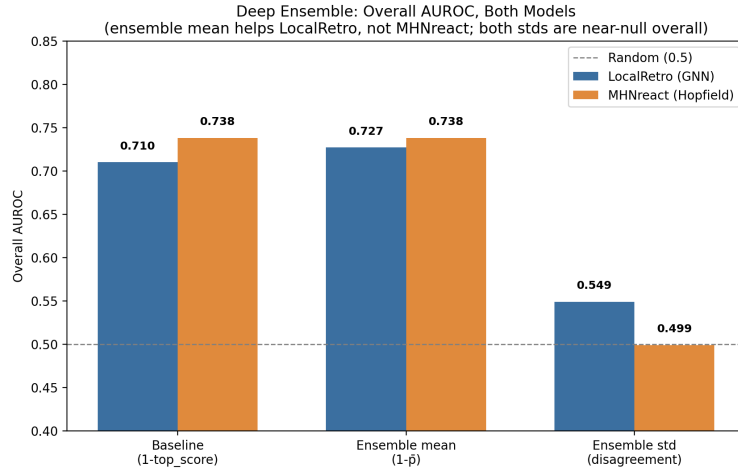


Figure 26: Deep ensemble overall (non-tiered) AUROC, both models. Ensemble mean helps LocalRetro but not MHNreact; both ensemble-std signals hover near random overall, which is why the very-rare-tier breakdown below is the figure that carries the argument.

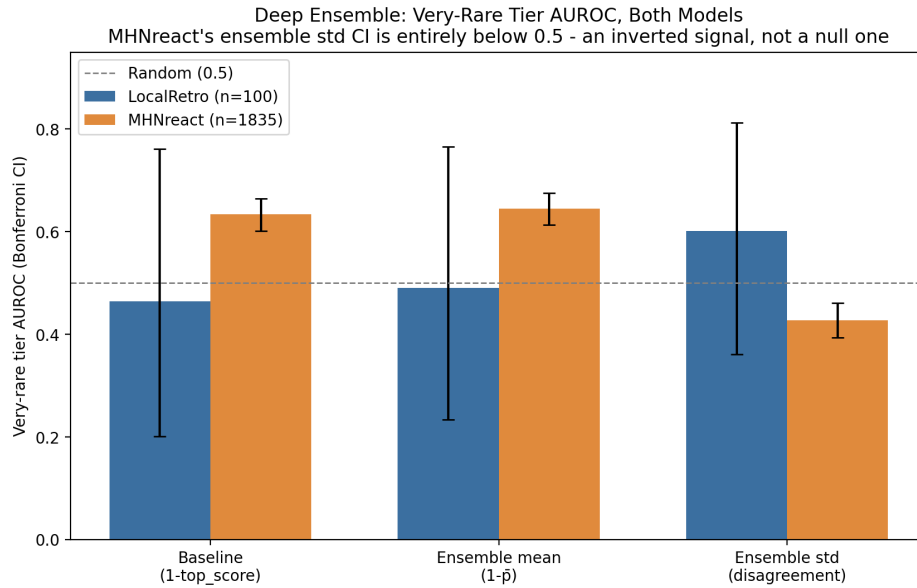


Figure 27: Very-rare tier AUROC with Bonferroni CIs, both models. LocalRetro's ensemble std CI spans 0.5 (a clean null). MHNreact's ensemble std CI sits entirely below 0.5, a statistically real inversion, not a wider, less-certain null.

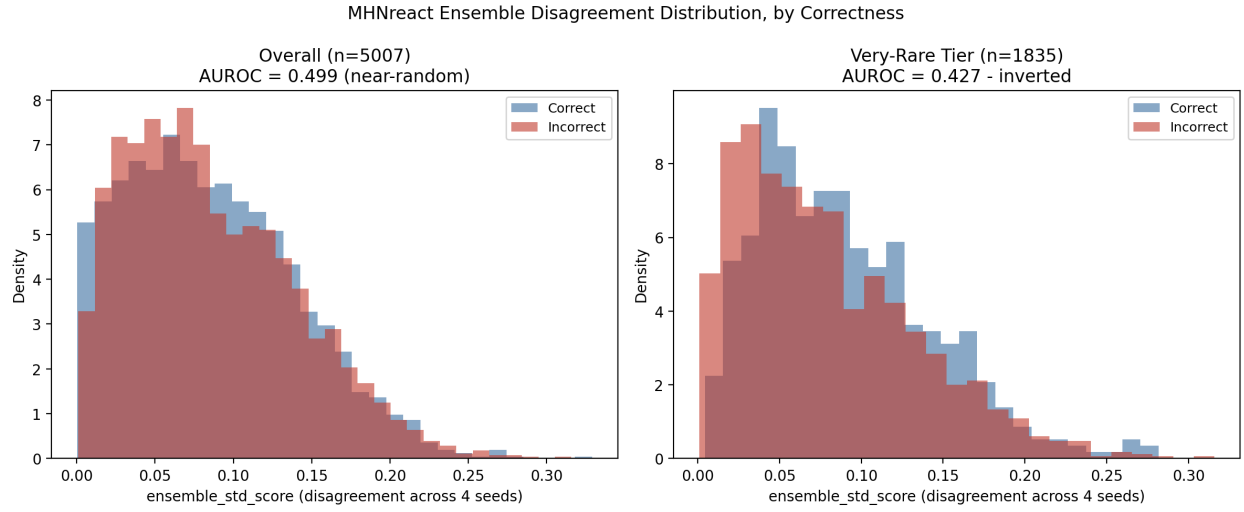


Figure 28: MHNreact ensemble disagreement distribution by correctness, overall vs. very-rare tier. The rightward shift of the “correct” distribution relative to “incorrect” in the very-rare panel is the visual signature of the inversion.

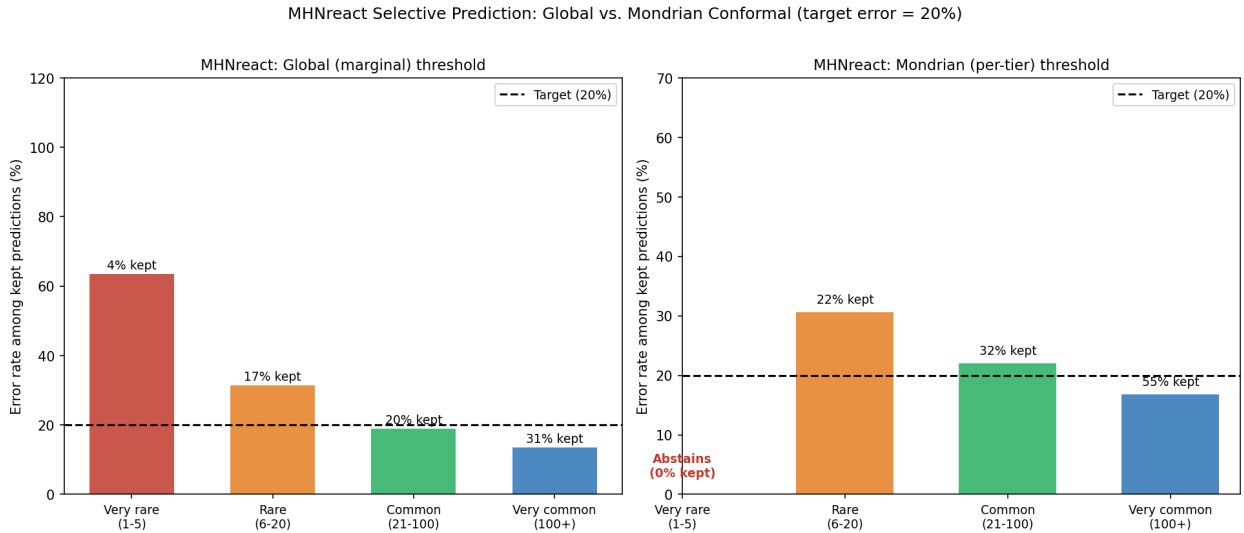


Figure 29: MHNreact selective prediction under a 20% target error rate, same harness as LocalRetro (Figure 4). *Left*: global threshold. *Right*: Mondrian per-tier thresholds.

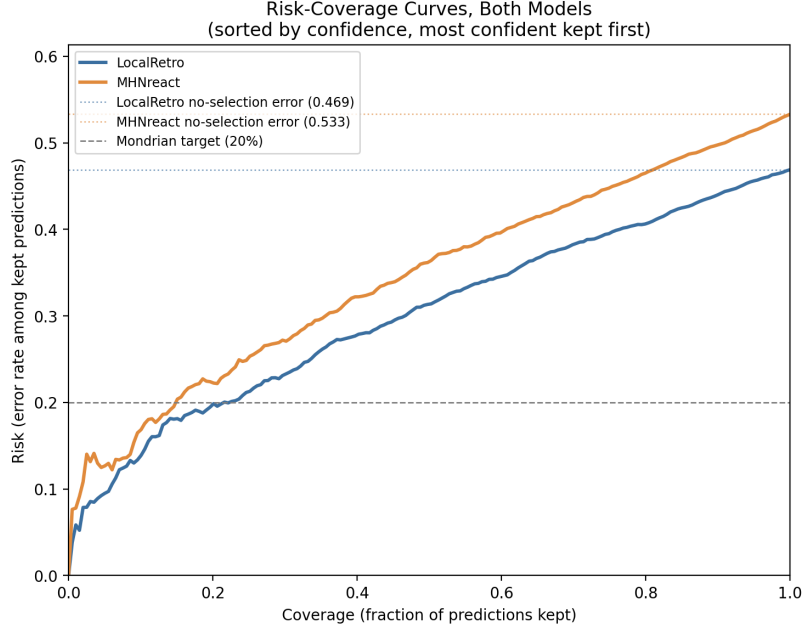


Figure 30: Overall risk-coverage curves, both models (sorted by confidence, most-confident-first, no tier stratification).

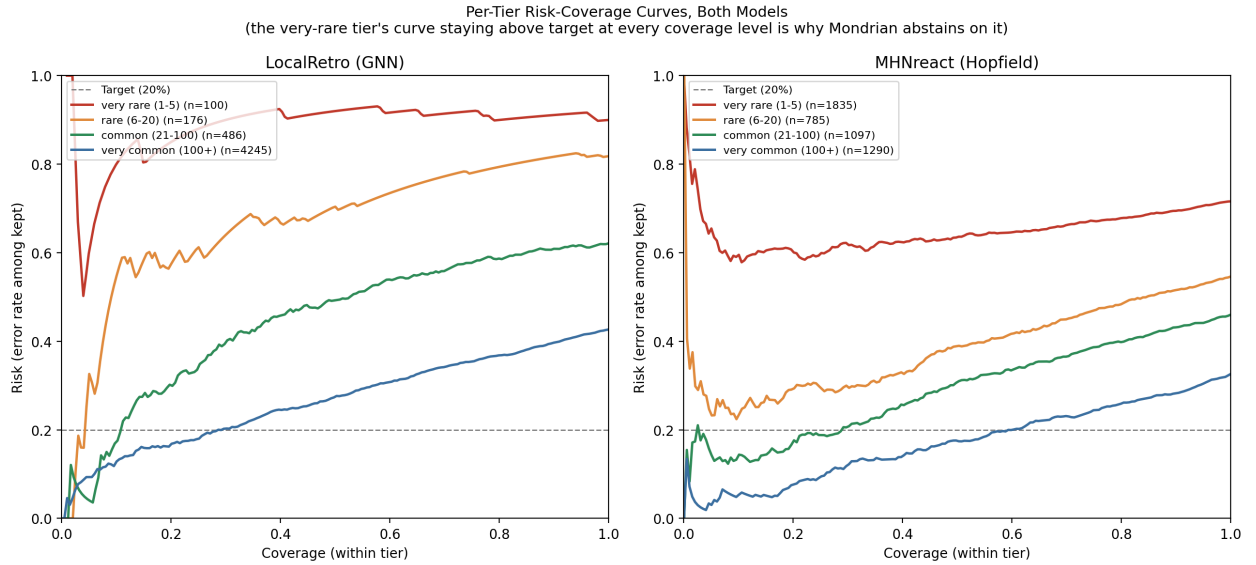


Figure 31: Per-tier risk-coverage curves, both models. LocalRetro’s very-rare-tier curve (red, left panel) never drops to the 20% target line even at very low coverage. MHNreact’s tiers (right panel) are more tightly clustered and each eventually crosses the target at some coverage level, though the rare tier crosses it only at low coverage, consistent with the finite-sample slack documented in Section 8.